



OPEN

Improving frequency stability in grid-forming inverters with adaptive model predictive control and novel COA-jDE optimized reinforcement learning

Muhammad Zubair Yameen^{1,2}✉, Zhigang Lu^{1,2}, Fayez F. M. El-Sousy³, Waqar Younis^{1,2}, Baqar Ali Zardari⁴ & Abdul Khaliq Junejo⁵

The increasing utilization of renewable energy sources in low-inertia power systems demands advanced control strategies for grid-forming inverters (GFMs). Conventional Model Predictive Control (MPC) methods, which depend on static models and predefined boundaries, often struggle to preserve frequency stability in dynamic grid conditions. This research presents an Adaptive Model Predictive Control (AMPC) framework to enhance GFM performance in Virtual Synchronous Machine (VSM) mode, ensuring robust frequency stability under uncertainties. The primary issue addressed is the inefficiency of traditional MPC in adapting to dynamic grid conditions. To resolve this, the AMPC framework combines offline reinforcement learning for parameter tuning with online MPC using soft constraints. The offline phase employs a novel Hybrid Crayfish Optimization and Self-Adaptive Differential Evolution Algorithm (COA-jDE) to minimize the cost function $U_{offline}$, deriving optimal control parameters (Q , R) before real-time deployment. This process, termed cost function minimization using COA-jDE in a reinforcement learning framework, enhances GFM performance by adaptively adjusting virtual inertia and damping. Simulations on a 16MW wind-powered DFIG microgrid demonstrate that AMPC outperforms traditional MPC and VSM methods during grid disturbances, symmetrical faults, islanding, and load shifts. Furthermore, AMPC is computationally efficient compared to conventional reinforcement learning techniques, as adaptation is restricted to offline tuning. The framework not only improves compliance with grid codes (e.g., GC0137, IEEE 1547) but also provides a flexible, resilient control strategy for modern low-inertia grids.

Keywords Model Predictive Control, Crayfish Optimization algorithm, Microgrid, Frequency, DFIG, Virtual Synchronous Machine, Differential Evolution algorithm

Abbreviations

GFM	Grid Forming Inverter
MPC	Model Predictive Control
AMPC	Adaptive Model Predictive Control
VSM	Virtual Synchronous Machine
MG	Microgrid
COA	Crayfish Optimization Algorithm
jDE	Self-Adapting Differential Evolution
SMC	Sliding Mode Control
RES	Renewable Energy Sources

¹School of Electrical Engineering, Yanshan University, Qinhuangdao 066004, China. ²Key Laboratory of Power Electronics for Energy Conservation and Drive Control of Hebei Province, Yanshan University, Qinhuangdao 066004, China. ³Department of Electrical Engineering, College of Engineering, Prince Sattam bin Abdulaziz University, Al Kharj 16273, Saudi Arabia. ⁴Department of Information Technology Quaid e Awam University Nawabshah, Nawabshah, Pakistan. ⁵State Key Laboratory of Advanced Electromagnetic Technology Huazhong University of Science and Technology, Wuhan, China. ✉email: zubbairyamin@gmail.com

VSI	Voltage Source Inverter
DFIG	Doubly Fed Induction Generator
DER	Distributed Energy Resources
SG	Synchronous Generator
IIRES	Inverter-Interfaced renewable energy sources
GFL	Grid Following Inverter
VSG	Virtual Synchronous Generator
PCC	Point of Common Coupling
CR	Crossover Rate
F	Mutation factor
PV	Photovoltaic
PMSG	Permanent Magnet Synchronous Generator
IBR	Inverter Based Resources
ESS	Energy Storage Systems

The increasing integration of inverter-interfaced renewable energy sources (IIRES) has fundamentally changed the dynamics of current power systems, resulting in a significant decrease in grid inertia, which is essential for preserving frequency stability¹. Unlike typical synchronous generators (SG), which provide inertia through rotating masses, inverter-based resources (IBRs) are mechanically decoupled from the grid, restricting their ability to immediately respond against frequency disturbances². As the share of renewable energy sources such as wind and solar power increases, power systems encounter new issues in maintaining stability and reliability³. The transition from centralized to decentralized generation worsens this problem, as microgrids (MG), which include distributed generation (DG), energy storage systems (ESS), and local loads, are used more and more to incorporate renewable energy⁴. While microgrids provide promising solutions to energy and environmental challenges, they raise new stability concerns, particularly because synchronous machines are replaced by converter-based RESs, which lack the inertia and damping required to stabilize frequency and damp system dynamics. As a result, the reduced system inertia causes larger frequency variations, posing significant threats to grid reliability⁵.

To tackle these issues, advanced control techniques have become essential for maintaining grid stability. Despite their widespread use, conventional Grid-Following Inverters (GFL) frequently underperform in dynamic grid situations, resulting in frequency and voltage instability⁶. To overcome this, Grid-Forming Inverters (GFM) have emerged as the key technology⁷. GFMs actively control grid frequency and voltage by simulating the inertial response of synchronous generators through the Virtual Synchronous Machine (VSM) structure. Unlike conventional converters, which lack synthetic inertia and damping critical for mitigating transient instabilities (e.g., rapid frequency deviations during faults), VSM-based GFMs emulate electromechanical dynamics to provide inertia, droop response, and fault ride-through capabilities mandated by modern grid codes (e.g., IEEE 1547-2018, GC0137). This ensures system-wide stability during disturbances while aligning with traditional grid operational paradigms.

By integrating VSM with the Adaptive Model Predictive Control (AMPC) framework, we synergize synchronous machine robustness with real-time optimization, achieving adaptive stability without sacrificing grid code compliance. GFMs further excel in low-inertia or islanded grids by autonomously regulating active and reactive power with rapid response times inherent to inverter-based systems⁸. These capabilities enhance system durability across diverse operational scenarios, from steady-state conditions to severe transients.

Recent studies have underscored the growing importance of GFMs, particularly in Doubly Fed Induction Generator (DFIG)-powered wind farms. The complex interactions between these intermittent generation systems and the grid often overwhelm traditional control mechanisms⁹. GFMs are able to not only improve voltage and frequency regulation while satisfying regulatory standards, but they also provide critical frequency and voltage references, making them vital for maintaining grid support in inverter dominated grids, particularly during islanding events.

Grid-Forming Inverters in Virtual Synchronous Machine (VSM) mode have become a pivotal technology for frequency stability and increasing damping in power systems with high renewable energy penetration¹⁰. Several control schemes for GFMs have been devised to increase frequency stability, each with benefits and drawbacks^{11,12}. This section examines recent research on these strategies, showing both their benefits and downsides.

In¹³, typical VSM control for GFMs simulates the behavior of a synchronous machine by providing fixed virtual inertia. Although this strategy works well for moderate interruptions, it is ineffective for low-inertia grids that undergo rapid changes. Furthermore, fixed inertia lowers system flexibility, causing slower reactions against significant frequency changes. Sliding Mode Control (SMC) for VSM-based GFMs, as implemented in¹⁴, improved frequency profile under uncertain conditions but caused issues such as chattering, which could reduce performance in real-world applications. Consensus-based control for decentralized systems increased inverter coordination, although communication delays during rapid load shifts reduced frequency stability¹⁵. Fuzzy logic control, as outlined in¹⁶, reacted effectively to changing grid conditions but relied heavily on expert knowledge for rule development, limiting its scalability across various scenarios.

A hierarchical control approach combining centralized control with VSM was proposed in¹⁷, improving stability during islanding and reconnection events, but its complexity limited its real-time performance in large networks. In¹⁸, the State-Space control approach effectively regulated frequency in low-inertia grids but was sensitive to parameter estimation errors, making it less reliable under real-world conditions. H-infinity control for VSM-based GFMs in¹⁹ provided robustness against perturbations but had slower response times, limiting its effectiveness during rapid load fluctuations.

Advanced VSM control techniques combining machine learning were used in²⁰ to continuously change GFM control settings, increasing frequency regulation flexibility, however, real-time applicability was limited by training delays. Nonlinear control approaches, as investigated in²¹, enhanced performance during rapid frequency shifts, but their degree of complexity in real-time tuning limited their application. Conventional Model Predictive Control (MPC) for VSM, as discussed in²², demonstrated potential for stabilizing low-inertia grids. However, it encountered challenges with adaptability during disruptions due to its dependence on static models. To improve MPC soft constraints were incorporated into MPC in²³ to enhance flexibility, but key challenges still persisted during serious events, especially during huge frequency changes. Adaptive MPC for GFMs developed in²⁴ enables real-time adjustments according to grid conditions. Although this strategy improved frequency stability, its computational complexity posed considerable challenges, especially in systems with distributed energy resources (DERs). Dynamically adjusted virtual inertia in adaptive VSM control, as proposed in²⁵, performed well during frequency disturbances but required precise real-time measurements, making implementation difficult. Reinforcement learning (RL) for GFM control improves frequency response but faces computational bottlenecks and catastrophic forgetting during dynamic grid interactions. Recent advancements like the proximal brain-inspired policy optimization (PBPO)²⁶ address these issues by emulating cognitive processes, enabling robust frequency regulation in grids with high renewable penetration. Complementing this, digital twin frameworks²⁷ ensure safe agent deployment via offline training and centralized-decentralized control, aligning with modern grid code compliance while mitigating operational risks.

Despite advances in conventional and adaptive VSM control systems, these existing VSM control strategies suffer from unique limitations. Static models and preset limits in methods such as conventional MPC and VSM control reduce efficacy in dynamic situations, whereas adaptive and learning-based approaches frequently encounter high complexity and computational burdens. This study tackles these limitations by introducing a novel Adaptive Model Predictive Control (AMPC) technique that combines precision, real-time performance, and scalability, resulting in an effective solution for improving frequency stability in low-inertia microgrids.

A comparative study of several control techniques and their applications to GFMs for frequency stability is shown in Table 1. Every cited study highlights different approaches, including control systems, techniques, and the related benefits and drawbacks. Although typically lacking in adaptability or computing efficiency, approaches such as VSM control, conventional Model Predictive Control (MPC), and adaptive control with reinforcement learning have shown promise in achieving particular degrees of frequency control. Conversely, the final row presents the proposed AMPC technique, designed to address these limitations. The AMPC methodology, which combines offline reinforcement learning with the novel hybrid Crayfish and Self-Adaptive Differential Evolution (COA-jDE) algorithm and incorporates slack variables for constraint handling in online operation, outperforms previous solutions. In low-inertia power grids, AMPC specifically offers improved frequency regulation, increased grid adaptability, and reduced computational burden, making it a more reliable and effective option. This table illustrates how the proposed AMPC method outperforms existing techniques.

Table 2 summarizes the essential technical requirements and performance standards for grid-forming inverters (GFM), including voltage and frequency ride-through capabilities, reactive power support, and the critical ability to simulate inertia. These criteria, developed by GC0137 and IEEE 1547, are critical to ensuring reliable operation in grid-tied and islanded modes^{30,31}.

The proposed Adaptive MPC for VSM assures complete compliance with all regulatory standards, particularly the precise emulation of virtual inertia as stipulated by GC0137. This technique improves grid stability, even under tough grid conditions, by allowing the GFM to dynamically adjust to voltage and frequency variances.

In addition to introducing the novel Adaptive Model Predictive Control (AMPC) method in conjunction with the hybrid optimization technique COA-jDE, there is a significant research gap in using microgrids (MGs) for frequency stabilization rather than solely relying on distributed generators (DGs) or individual wind turbine generators. The subsequent section explores the distinctive contributions of this research.

Motivation and contribution

The fundamental goal for this study is to address limitations in existing strategies for stabilizing low-inertia microgrids (MGs) and decrease the response time of inverters, particularly those interfaced with inverters, such as DFIG-based MGs. Because of the reduced inertia provided by renewable energy sources, these microgrids are especially susceptible to frequency excursions and grid disturbances. By providing a reliable solution, this work outperforms previous techniques in preserving frequency stability under a range of operating circumstances, guaranteeing stable operation of inverter-interfaced MGs in both grid-connected and islanded modes. This novel control strategy further favors low computing complexity and minimum real-time computational cost, making it ideal for practical use. Compliance with the GC0137 grid code requirements facilitates its incorporation into current, renewable-focused power systems.

This paper offers the following contributions:

- **Introducing a new AMPC framework:** The proposed AMPC technique uses the COA-jDE algorithm for offline reinforcement learning, improving the policy prior to its real-time application, which improves adaptability and minimizes computational demand in online operations. Offline reinforcement learning aids in the development of robust control rules by allowing for lengthy training while preserving real-time performance.
- **Implementation of soft constraints:** Soft constraints for frequency and voltage ensure stable operation during grid disruptions by balancing stability and control flexibility, allowing for smooth deviation handling.
- **Real-time adaptability to frequency deviations:** AMPC effectively stabilizes system frequency by dynamically modifying its active power output in response to frequency deviations, thus assuring frequency remains within acceptable limits.

Ref	Controlling Equipment	Controlling Technique	Strengths	Shortcomings
13	PLL enabled GFM backed by ESS	Robust H_{∞} Control with VSM	Enhanced frequency stability in low-inertia microgrids, Robust against parameter uncertainties (system inertia, damping, and PLL effects), Superior performance over conventional PI-based control	Increased complexity in controller design due to robust H_{∞} approach, High controller order (9th order) before reduction, which adds computational burden, Limited focus on real-time implementation challenges or hardware constraints.
14	LCL-filter interfaced GFM	Higher Order Sliding Mode Control (HO-SMC)	Enhances frequency response in weak grids, Offers robustness against parameter variations and disturbances, Outperforms classical vector control (CVC) in terms of transient stability and fault ride-through capability.	Complex control design increases implementation challenges, Higher switching rates may lead to issues if not managed properly, Scalability is not addressed, Proposed control is only tested for grid-connected mode only
15	GFM and Synchronous Generators	Hierarchical two-layer control (primary droop-based, secondary consensus-based) with safety filters.	Safe transient frequency control, Minimizes frequency deviations, Distributed control improves scalability and resilience.	Scalability limited by droop gains and storage capacity, Requires high communication and coordination for consensus control, Real-time implementation can be computationally expensive.
28	GFM combined with BESS and wind turbines	Self-adaptive virtual inertia control using fuzzy logic	Enhances frequency stability in low-inertia microgrids with high renewable penetration, Automatically adjusts virtual inertia in real-time based on system conditions, Mitigates frequency oscillations caused by renewable energy disturbances.	Relies on communication links, vulnerable to delays/failures, Sensitive to noise in frequency measurements, Dependent on BESS capacity for performance, May not scale well with very high RES penetration or larger systems, Voltage stability is not as addressed as frequency.
17	GFM	Hierarchical control with primary, secondary, and tertiary control levels (droop control methods for primary control in GFMs)	Provides a stable reference for voltage and frequency in island mode, High reliability with distributed control in grid-connected and island modes, Improves system flexibility for grid-connected and islanding transitions.	Vulnerable to network delays or failures in communication-based controls, Scalability issues due to centralized control and complex hierarchical adjustments, High implementation and maintenance costs from communication infrastructure, Slower real-time adaptability due to potential communication delays, Risk of uneven load sharing with droop control in communication-free methods.
18	GFM	State-Space modeling framework with small-signal stability analysis to stabilize the GFMs (VSM emulation)	Enhances grid stability by emulating inertia without the need for energy storage, Provides improved frequency nadir and RoCoF under disturbances, Compatible with existing grid-forming converters, enabling distributed and flexible inertia control,	Relies heavily on precise tuning of the lead compensator, making setup complex, Limited scalability due to configuration-specific tuning requirements, Increased design complexity from integration of multiple control layers, Stability could be compromised if system conditions vary significantly, Potential difficulties in maintaining control effectiveness across diverse grid setups, Higher implementation and maintenance costs due to control complexity
19	LCL-filter interfaced GFM	H_{∞} optimal control for GFM	Achieves robust stability and performance even under variable grid conditions, Decentralized control with small-gain theorem for stability in multi-converter systems, Reduces the need for trial-and-error parameter tuning by providing systematic control design.	Requires precise tuning of H_{∞} weighting functions, increasing design complexity, Limited scalability; configuration-specific tuning may not adapt to larger or more complex systems, Static controller design may struggle with large, sudden disturbances, High implementation cost due to complex decentralized control structure,
20	Voltage Source Inverter (VSI)	Model-free predictive control (MF-PC) with an autoregressive model (ARX) for estimation	Highly resilient to model inaccuracies as it Predicts output without needing system model parameters, Quick response to transient events and stable operation in varied grid conditions, Maintains low total harmonic distortion (THD) even with parameter mismatches.	ARX model parameters need frequent adjustments, This control technique has a higher computational load than traditional controllers, Sensitivity to measurement noise (in frequency and voltages) can impact stability, Performance depends heavily on the accuracy of real-time data
21	Grid-forming voltage source converter (VSC) with a virtual synchronous generator (VSG) control	Nonlinear decoupling method implemented in VSG	Enhances stability by dynamically adjusting damping and inertia, Effectively maintains frequency stability and synchronizes with grid.	High computational complexity limits real-time application, Sensitive to parameter variations, requiring precise tuning, Limited scalability for large or multi-converter systems, Requires accurate real-time data, making it vulnerable to communication delays
22	GFM is implemented in VSI	Model predictive control (MPC)	Enhances frequency stability by coordinating load and inverter response, Incorporates real-time constraints on rate of change of frequency (RoCoF) and load ramping, Effectively adapts to load variations and disturbances	Uses static model and preset limits, reducing flexibility under unexpected conditions, Relies on fast, accurate data transmission, sensitive to delays, Frequent parameter tuning is required to maintain performance, Proposed methodology has limited focus on voltage stability under significant disturbances,
Continued				

Ref	Controlling Equipment	Controlling Technique	Strengths	Shortcomings
23	Voltage Source Converters (VSC) in DER units	Decentralized Model Predictive Control (MPC) applied within VSCs based on safety constraints	Ensures stability and safety constraints during disturbances, Manages transitions smoothly between islanded and grid-connected modes, Effective at maintaining power quality under load changes	High computational demands limit scalability with more DER units, Sensitive to inaccuracies in local measurements and state estimation, Complexity in parameter tuning for varied operating conditions.
25	GFM based Virtual Synchronous Generator	Adaptive Inertia and Damping Coordination (AIDC) Control using a VSG framework	Enhances both frequency stability and transient synchronization stability, Adaptive inertia and damping improve dynamic response under fault conditions, Broader attraction region compared to traditional methods, improving fault tolerance.	High complexity in designing and tuning adaptive inertia and damping control, Limited scalability; difficult to extend to larger or multi-area systems with diverse dynamics, Increased computational burden due to continuous real-time adjustments, Dependency on precise tuning; performance may degrade under large disturbances
29	GFM control implemented on Battery Energy Storage System (BESS)	Rule-based adaptive control strategy for virtual synchronous machine (VSM)	Improves frequency stability in islanded systems under varying renewable penetration, Adaptively adjusts inertia and damping to respond to disturbances and faults, Supports smooth recovery after faults and reduces the risk of frequency excursions	Increased computational load due to continuous real-time adjustments, Requires precise parameter tuning, which may not be effective in systems with unpredictable dynamics, High design complexity in setting adaptive thresholds for various network states, Potential delays in adaptive response under rapid, severe disturbances
This Study	Grid Forming Inverter (GFI) implemented in Virtual Synchronous Machine (VSM) model	Adaptive Model Predictive Control (AMPC) combining offline reinforcement learning with the novel COA-jDE algorithm, and real-time MPC incorporating slack variables for effective constraint handling during operation.	AMPC offers enhanced frequency stability with reduced computational complexity and improved adaptability to changing grid conditions, Reduced reliance on static models or predefined constraints, Improved responsiveness to sudden frequency deviations, High accuracy in dynamic environments with complex disturbances, Supports scalability across larger and more diverse power networks, Comparative analysis is also done.	

Table 1. Detailed overview of control strategies in existing literature for GFM frequency stabilization.

- **Performance demonstration under key scenarios:** The AMPC approach is evaluated using scenarios such as load changes, reference phase angle shift, and islanding, indicating its ability to improve the PCC frequency profile under varied conditions.
- **Enhanced synchronization in grid-connected mode:** The suggested GFM control responds faster in grid-connected mode than traditional MPC and VSM control, demonstrating improved synchronization capabilities.
- **Autonomous operation in island mode:** In island mode, the microgrid delivers frequency and voltage references autonomously, demonstrating its ability to operate reliably on its own.
- **Scalability through microgrid implementation:** Scalability is addressed by using a microgrid, which allows for effective expansion while preserving stable operation, as opposed to depending just on individual distributed generators or turbines.
- **Robust low-voltage ride-through (LVRT) capability:** The GFM control demonstrated improved resilience in grid-connected mode when tested with an extended LLLG fault period of 2 seconds (in addition to the 1-second requirement of GC0137).

This study presents a novel Adaptive Model Predictive Control (AMPC) architecture for GFMs that overcomes the constraints of standard control strategies. By combining reinforcement learning and real-time optimization, the AMPC framework improves frequency stability in low-inertia power systems. Using the COA-jDE technique for offline policy optimization enables it to effectively manage a wide range of disturbances in online mode while maintaining grid code compliance. Furthermore, the inclusion of slack variables in the cost function enables effective constraint handling during real-time operations, resulting in a practical and scalable solution for modern grid applications.

The remaining sections of this paper are organized as follows: Section [Fundamental Equations of Grid-Forming Inverter for Virtual Synchronous Machine \(VSM\) Control](#) provides an in-depth overview of the fundamental GFM control equations in VSM mode. Section [Proposed AMPC with Reinforcement Learning in VSM Mode](#) introduces the proposed AMPC control in VSM mode, detailing its integration with reinforcement learning. Section [Fault Current Control Strategies in Grid-Forming Inverters](#) then outlines the fault current control strategy applied within this GFM framework. In Sections [Crayfish Optimization Algorithm \(COA\)](#), [Differential Evolution Optimization Algorithm \(DE\)](#), and [Self-Adaptive Differential Evolution \(jDE\)](#), the Crayfish Optimization Algorithm, Differential Evolution, and Self-Adaptive Differential Evolution (jDE) algorithms are discussed comprehensively. The hybrid COA-jDE algorithm, developed to minimize cost function in offline reinforcement learning, is presented in Section [Hybrid COA-jDE for Cost Function Minimization in Offline Reinforcement Learning](#), with a detailed explanation of its operational steps. The test system and its configuration are introduced in Section [System Configuration](#). Finally, Section [Results and Discussion](#) provides

Parameter	Voltage/Frequency Deviation	Ride-Through/Response Duration	Tolerance Range	Standard	Description
Nominal Voltage	100% (e.g., 11kV)	Continuous	$\pm 0.5\%$	GC0137	GFM operate continuously at nominal voltage under normal conditions.
Voltage Sag (Moderate)	90% to 110% of nominal	Continuous	$\pm 10\%$	GC0137, IEEE 1547	GFM must handle minor voltage variations without disconnecting.
Voltage Sag (Severe)	50% of nominal voltage	250 milliseconds	-50%	GC0137	GFM must stay connected during severe voltage dips to assist with grid recovery.
Voltage Collapse	0% to 50% of nominal voltage	100 milliseconds	-100% to -50%	GC0137	GFM must remain connected momentarily even in extreme voltage collapses.
Overvoltage	$> 110\%$ of nominal voltage	Continuous	10%	GC0137	GFM can handle overvoltage situations by maintaining system stability.
Frequency Support	$\pm 2\%$ from nominal (50 Hz/60 Hz)	Continuous	$\pm 2\%$	IEEE 1547	GFM provide active power support to stabilize frequency deviations.
Low Frequency Ride-Through	< 47.5 Hz (for 50 Hz systems)	0.5 seconds	-5%	GC0137, IEEE 1547	GFM should remain operational for short frequency drops before disconnection.
High Frequency Ride-Through	> 52.0 Hz (for 50 Hz systems)	0.5 seconds	4%	GC0137, IEEE 1547	GFM should remain connected for brief high-frequency events.
Inertia Emulation	Emulate inertia equivalent to $0.1 - 0.3$ s H	Continuous (VSM control mode)	n/a	GC0137	GFM must emulate inertia equivalent to $0.1-0.3$ seconds H, to stabilize the system during frequency deviations.
Active Power Frequency Response	Power output variation ($f \pm 2\%$)	Continuous	$\pm 2\%$ frequency variation	IEEE 1547	GFM adjusts active power based on frequency deviations to ensure system balance.
Reactive Power Voltage Response	VAR output variation ($V \pm 5\%$)	Continuous	$\pm 5\%$ voltage variation	IEEE 1547	GFM adjusts reactive power to stabilize voltage within the network.
	Down to 0% of nominal voltage	150 ms (min) to 1 second (max)	n/a	GC0137, IEEE 1547	GFM should ride through voltage sags and remain operational to support the grid.
	0.8 to 0.5 pu	90 seconds or more	Remain connected	GC0137	GFM must stay connected and support voltage recovery when the voltage drops between 0.8 and 0.5 pu for 90 seconds or more.
Voltage Ride-Through (VRT)	0.5 to 0 pu	500 ms	Disconnection allowed	GC0137	GFM is required to remain connected for 500 milliseconds when the voltage drops below 0.5 pu, after which disconnection is permitted.

Table 2. Key standards and requirements for Grid-Forming Inverters (GFM) from GC0137 and IEEE 1547^{30,31}.

an analysis of the results, while Section [Conclusions and future research](#) concludes the paper and highlights areas for future research.

Fundamental equations of Grid-Forming inverter for virtual synchronous machine (VSM) control

A grid-forming inverter operating in Virtual Synchronous Machine (VSM) mode emulates the behavior of a synchronous generator by establishing the grid's reference voltage and frequency. In doing so, it contributes virtual inertia and damping to stabilize frequency and voltage while facilitating power sharing among inverter-based resources.

Fig. 1 shows the control diagram for typical GFM control in VSM mode. The following equations capture the dynamic behavior of the inverter when configured as a VSM.

The dynamic response is governed by the swing equation:

$$2H \frac{d\omega}{dt} = P_m - P_e - D(\omega - \omega_0) \quad (1)$$

Here, H simulates rotational inertia, dictating the inverter's response to frequency deviations ($\frac{d\omega}{dt}$). P_m (reference power) and P_e (measured output power) drive frequency dynamics, while $D(\omega - \omega_0)$ stabilizes deviations from the nominal frequency ω_0 .

The active power output (P_e) depends on the voltage magnitudes (V_{inv} , V_{grid}) and phase angle difference (δ) between the inverter and grid:

$$P_e = \frac{V_{inv} \cdot V_{grid}}{X} \cdot \sin(\delta) \quad (2)$$

X represents the equivalent reactance at the point of common coupling (PCC). The sine term ($\sin(\delta)$) models the power-angle relationship, akin to synchronous generators, ensuring active power injection aligns with grid synchronization.

The phase angle (δ) evolves proportionally to the frequency deviation ($\omega - \omega_0$) via Eq.(3):

$$\frac{d\delta}{dt} = \omega - \omega_0 \quad (3)$$

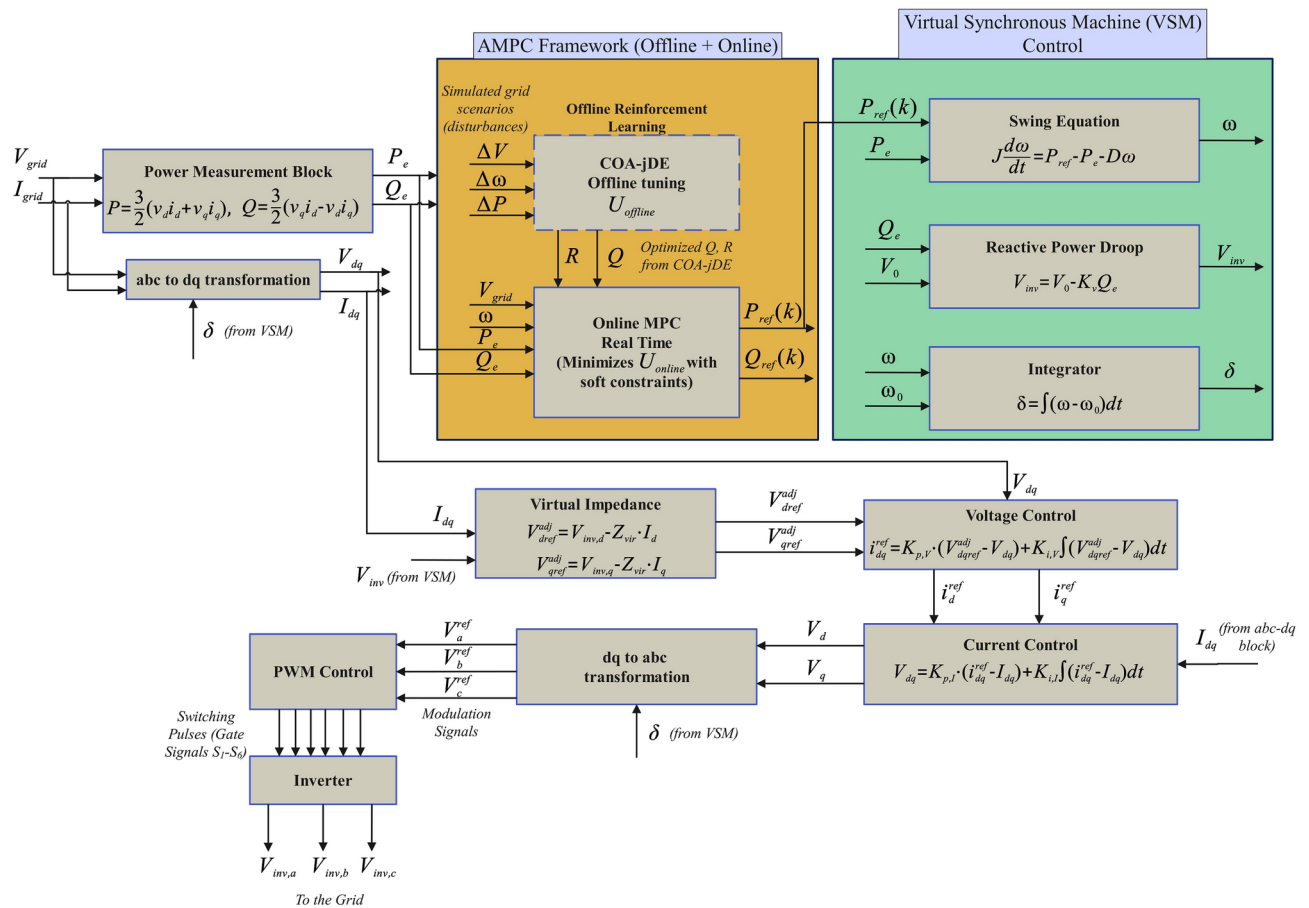


Fig. 1. VSM based GFM control³².

This equation ensures the inverter's internal phase remains synchronized with the grid, critical for stability during transients.

Voltage regulation follows a reactive power droop control (Eq.(4)):

$$V_{inv} = V_0 - K_v Q_e \quad (4)$$

V_0 is the nominal voltage, Q_e the measured reactive power, and K_v the droop coefficient. This allows decentralized voltage management, enabling multiple inverters to share reactive power without communication.

The reactive power output (Q_e) is governed by Eq.(5):

$$Q_e = \frac{V_{inv}(V_{inv} - V_{grid} \cdot \cos(\delta))}{X} \quad (5)$$

This equation highlights the dependency of Q_e on voltage differences and phase alignment, critical for voltage stability in IBR-dominated grids.

The equations above form the foundation of the GFM VSM control architecture (Fig. 1), which emulates synchronous machine dynamics through a hierarchical control structure. Central to this is the swing equation (1), which governs the dynamic frequency (ω) by balancing the reference power (P_{ref}) and measured electrical power (P_e). Here, ω_0 denotes the nominal grid frequency (e.g., 50Hz), while ω represents the inverter's internal frequency, deviating during transients. The phase angle (δ) is computed by integrating $\omega - \omega_0$ (Eq. (3)), ensuring synchronization with the grid.

Reactive power regulation follows the voltage droop control (Eq. (4)), where V_0 is the nominal voltage, Q_e is the measured reactive power, and K_v is the droop coefficient. The voltage reference V_{inv} is adjusted by a virtual impedance (Z_{vir}) to account for inverter current (I_{dq}), yielding the adjusted references V_{dref}^{adj} and V_{qref}^{adj} .

These references are tracked by nested voltage and current control loops, generating modulation voltages (V_d and V_q) in the dq -frame. These signals are transformed to the abc -frame using δ and fed to the PWM block to produce switching signals ($S_1 - S_6$) for the inverter. The inverter output is filtered through an LCL filter, producing stable sinusoidal voltages ($V_{grid,abc}$) at the PCC. This architecture ensures grid synchronization, inertial response, and robust power sharing in inverter-dominated grids.

While these equations establish the core VSM dynamics, their practical implementation demands adaptive control to handle grid uncertainties. The next section introduces an Adaptive Model Predictive Control (AMPC) framework enhanced with reinforcement learning.

Proposed AMPC with reinforcement learning in VSM mode

In this part, we present a precise mathematical formulation of Adaptive Model Predictive Control (AMPC) for Grid-Forming Inverter (GFM) running in Virtual Synchronous Machine (VSM) mode. This method combines reinforcement learning for offline tuning with novel COA-jDE and Model Predictive Control (MPC) with soft constraints for real-time optimization. This VSM control improves the behavior of GFM by applying virtual inertia and damping intelligently to improve the grid's frequency stability.

Virtual Synchronous Machine (VSM) control equations

The VSM control is the primary component of our GFM system, replicating the behavior of typical synchronous machines using virtual inertia and damping. The swing equation, which balances the torque input with the system's inertia, damping, and output power, serves as the foundation for VSM behavior. The swing equation given in (1) can be rewritten as:

$$J \frac{d\omega}{dt} = P_{ref} - P_{out} - D\omega \quad (6)$$

Here, J represents the virtual inertia. P_{ref} refers to the reference power, which serves as a set point for the inverter. P_{out} represents the actual power output from the inverter. D is the damping coefficient that minimizes frequency oscillations by dissipating energy proportionate to frequency deviation and ω is the deviation in angular frequency from the typical grid frequency.

The VSM control attempts to balance the reference and output power while incorporating the effects of inertia and damping, hence stabilizing the frequency. The output of this eq (6) is sent into the GFM's control loop, which regulates the inverter response.

State-Space model incorporating VSM control and filter dynamics

To execute the AMPC method, the Grid-Forming Inverter's (GFM) dynamics must be described in a state-space model. This model combines VSM control with the dynamics of the LCL filter, which is widely employed in inverter systems to filter out high-frequency harmonics.

The state vector $x(k)$ comprises the important variables in the system: the frequency deviation $\omega(k)$, the inductor current $i_L(k)$, and the capacitor voltage $v_C(k)$.

$$x(k) = \begin{bmatrix} \omega(k) \\ i_L(k) \\ v_C(k) \end{bmatrix} \quad (7)$$

The general shape of the state-space model that shows how these states change over time is:

$$x(k+1) = Ax(k) + Bu(k) + Ed(k) \quad (8)$$

Where the state transition matrix A models the system's internal dynamics. B shows how the control input $u(k)$ influences the states. E collects external disturbances $d(k)$, including grid fluctuations. Adding this to our GFM system, the particular matrices are presented as:

$$A = \begin{bmatrix} 1 - \frac{DT_s}{J} & -\frac{T_s}{J} & 0 \\ \frac{T_s}{L} & 0 & -\frac{T_s}{L} \\ 0 & \frac{T_s}{C} & 1 \end{bmatrix}, \quad B = \begin{bmatrix} \frac{T_s}{J} \\ 0 \\ 0 \end{bmatrix}, \quad E = \begin{bmatrix} 0 \\ \frac{T_s}{L} \\ 0 \end{bmatrix} \quad (9)$$

Here, T_s is the sample time of the control system. The inductance and capacitance of the LCL filter is L and C respectively.

In AMPC, the control action $u(k)$ changes the system to minimize the cost function U over a certain prediction horizon. Since the GFM runs in VSM mode, $u(k)$ represents the inverter's modulation signals, which affect the output frequency and power injection. These signals are utilized to stabilize the frequency and regulate active/reactive power at the PCC.

The control input can be expressed as:

$$u(k) = [P_{ref}(k), Q_{ref}(k)] \quad (10)$$

where the reference active and reactive power setpoints generated by AMPC are $P_{ref}(k)$ and $Q_{ref}(k)$, respectively. The AMPC updates these references dynamically depending on expected future states, with the objective of lowering the cost function while meeting system limits.

The disturbance vector $d(k)$ represents fluctuations in external variables (voltage, frequency and power) as measured directly from the grid. To ensure PCC stability, the control method has to compensate for these disturbances.

The disturbance vector is defined as follows:

$$d(k) = [\Delta V(k), \Delta \omega(k), \Delta P(k)] \quad (11)$$

$\Delta V(k)$ represents the voltage deviation, $\Delta \omega(k)$ represents the frequency deviation, and $\Delta P(k)$ represents the power variation. These are real-time measurements of how the grid's conditions differ from their nominal values. This data is used by the AMPC to modify the control inputs.

The system's detailed state-space equations are:

$$\omega(k+1) = \left(1 - \frac{DT_s}{J}\right) \omega(k) - \frac{T_s}{J} i_L(k) + \frac{T_s}{J} P_{ref}(k) \quad (12)$$

$$i_L(k+1) = \frac{T_s}{L} \omega(k) - \frac{T_s}{L} v_C(k) \quad (13)$$

$$v_C(k+1) = \frac{T_s}{C} i_L(k) + v_C(k) \quad (14)$$

These equations describe how the system's frequency deviation, capacitor voltage, and inductor current change in response to control inputs and internal dynamics. This state-space model serves as the basis for the predictive control framework by combining both VSM and filter dynamics.

Prediction Model for frequency control over a horizon

MPC optimizes control inputs by forecasting system behavior over time. The prediction model for frequency deviation ω_k is expanded to three horizons, using dynamics from the state-space model.

The state-space model directly predicts frequency deviation one-step-ahead as given in Eq. (12). This equation predicts the frequency at the next time step, taking into account the current frequency deviation, inductor current, and control input P_{ref} .

To forecast the frequency two steps ahead, substitute $\omega(k+1)$ into itself:

$$\omega(k+2) = \left(1 - \frac{DT_s}{J}\right)^2 \omega(k) - \left(1 - \frac{DT_s}{J}\right) \frac{T_s}{J} i_L(k) + \frac{T_s}{J} P_{ref}(k+1) \quad (15)$$

This equation shows frequency development across two time steps, including the immediate influence of the present state and the delayed effect of future control inputs.

Adding three time steps to the forecast yields:

$$\omega(k+3) = \left(1 - \frac{DT_s}{J}\right)^3 \omega(k) - \left(1 - \frac{DT_s}{J}\right)^2 \frac{T_s}{J} i_L(k) + \frac{T_s}{J} P_{ref}(k+2) \quad (16)$$

This equation lets the controller forecast long-term control consequences. This multi-step prediction model lets the MPC optimize control inputs to minimize frequency deviations throughout the forecast horizon.

Adaptive Model Predictive Control (AMPC) framework

The Adaptive Model Predictive Control (AMPC) architecture combines offline reinforcement learning with real-time MPC to optimize control actions and modification of control parameters. AMPC aims to strike a compromise between reducing frequency deviations and meeting system restrictions while responding to changing operating circumstances.

- 1) **Offline reinforcement learning by using COA-jDE:** The offline tuning step in this control strategy involves minimizing the cost function $U_{offline}$. This function penalizes deviations in critical system variables (frequency in our case) from their reference value and control effort. This phase aims to optimize control parameters Q and R for use in the online MPC phase. Cost function minimization ensures the system behaves optimally by reducing frequency deviations and control effort

This offline reinforcement learning approach uses the Crayfish Optimization Algorithm in conjunction with Self-Adaptive Differential Evolution (COA-jDE). The proposed novel COA-jDE is discussed in detail in the Section 4. Q and R have been tuned during optimization so as to balance the trade-offs between maintaining frequency stability, voltage stability, and tracking power references. By assigning higher weights to frequency fluctuations and appropriate penalisation levels to real and reactive power, the control strategy effectively prioritizes critical aspects of system stability, ensuring that the GFM performs reliably in VSM mode.

The cost function $U_{offline}$ is defined as follows:

$$U_{offline} = \sum_{k=0}^{N_p} (\omega[k]^T Q \omega[k] + u[k]^T R u[k]) \quad (17)$$

$$\omega(k) = [\Delta f(k), \Delta V(k)]^T \quad (18)$$

$\omega(k)$ is a matrix that indicates the variation in frequency and voltage from their reference. $u(k)$ indicates the control effort at time step k . Q and R are the weight matrices tuned in this phase to balance the system's performance between reducing frequency deviations and control effort.

This offline phase follows a reinforcement learning (RL) paradigm, where minimizing the cost function $U_{offline}$ corresponds to maximizing a hypothetical reward $\rho_t = -U_{offline}$. The agent (control system) learns to optimize Q and R by iteratively reducing penalties, such as frequency deviations and control effort, in a simulated environment. Here, the “environment” is the simulated grid-forming inverter interacting with a local grid, and the “reward” in the RL context is defined as the negative of the cost function ($\rho_t = -U_{offline}$), reflecting reductions in frequency deviations and control effort. The novel COA-jDE method optimizes Q and R to minimize $U_{offline}$, thereby enhancing system performance.

A higher Q penalizes frequency and voltage deviations, assuring grid stability, whereas a higher R penalizes control operations, resulting in a more cautious control ($u[k]$) effort. The adjustment of these parameters provides a balanced performance in the real-time MPC phase, when the system is subjected to real-time disturbances and feedback.

Using simulation data during the offline phase exposes the system to a wide range of conceivable circumstances, preparing it to respond best during the online minimization phase.

In Fig. 2, the abscissa shows the number of iterations (episodes) and the ordinate indicates values of the cost function U .

As typical in reinforcement learning, the learning curve (reflecting cost minimization) is prone to non-monotonic behavior.

Initially, there are large fluctuations in the values of $U_{offline}$ which shows that the agent is trying out a number of actions and their results. Such fluctuations are normal at the early training stages because the agent has not figure out the best course of action and tries out several different policies.

With the increase in number of iterations, the agent starts to collect more information and thus there is an improvement in performance. This makes it possible for there to be a gradual decrease in the magnitude of oscillations. It is at this moment that the number of iterations approaches 3000 that the oscillations have minimized significantly, and the agent has stabled to the best policy.

The COA-jDE algorithm terminated optimization when $U_{offline}$ showed no significant improvement after 50–100 iterations. The initial value of $U_{offline}$ was ~ 5792.93 , which decreased to 303 after convergence, reflecting successful minimization.

To avoid trivial solutions ($Q = 0, R = 0$), we constrained diagonal entries of Q and R to $Q_{ii} \geq 1, R_{ii} \geq 0.1$. These bounds enforce stability requirements and ensure non-zero penalties on deviations and control effort.

The values of the Q and R matrices after reinforcement learning are provided by eq (19) and (20).

$$Q = \begin{bmatrix} 97.32 & 0 \\ 0 & 45.96 \end{bmatrix} \quad (19)$$

$$R = \begin{bmatrix} 0.96 & 0 \\ 0 & 0.47 \end{bmatrix} \quad (20)$$

The frequency deviation term in matrix Q is given a weight of 97.32. This shows that there is great attention to the frequency deviation it means that there is a high degree of control in the system in terms of stability and reliability since higher frequency fluctuations will affect the performance of power system negatively.

In contrast 45.96 represents the weight for voltage deviation which is lower than that of frequency. It indicates that voltage deviations are crucial but less critical than frequency deviations, as seen by the lower weight. This rationalization is consistent with the operational objectives of the system since the current stability parameters are well taken care of.

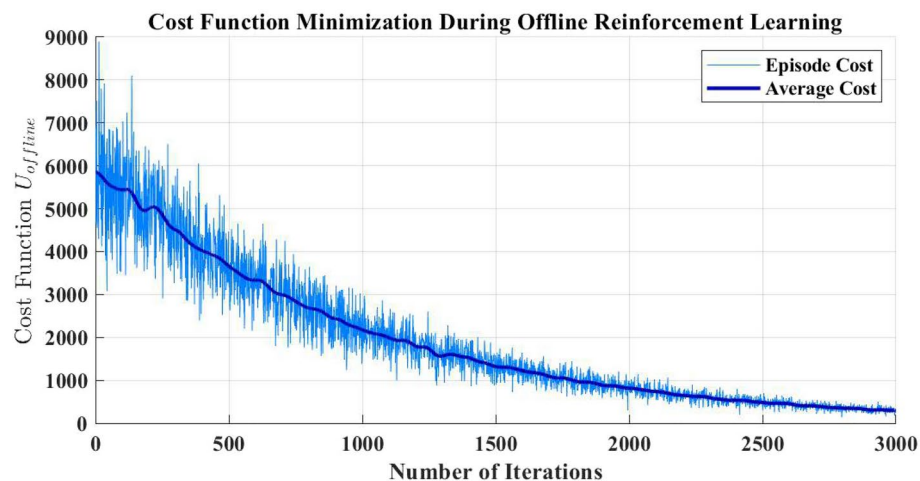


Fig. 2. Training curve for the cost function $U_{offline}$ during offline reinforcement learning. Lower values indicate reduced frequency deviations and control effort.

The value 0.96 for the power reference P_{ref} represents the standard level of penalization on deviations in power references. This weight indicates that, while correct power reference is essential, it is not as important as frequency variation.

The reactive power reference Q_{ref} has a smaller weight (0.47), indicating that deviations in reactive power are penalized less than those in real power. This reflects a balanced control strategy where reactive power variations, while less critical than frequency deviations, are still monitored and regulated to ensure grid stability.

Furthermore, to simplify the optimization problem, the off-diagonal values in the Q and R matrices are set to zero. This decouples frequency and voltage deviations, allowing focused analysis of their individual contributions to the cost function $U_{offline}$. This design choice enables systematic investigation of each variable's impact on control system performance without cross-interference from other components.

2) Online Model Predictive control (MPC) with soft constraints:

After the offline phase, the optimized Q and R values are used in the online MPC phase to minimize the cost function U_{online} , which now includes slack variables for soft constraints but with real-time measurements and system feedback. The online MPC phase employs quadratic programming to address the minimization problem. The system state ($\omega(k)$) is anticipated across a future horizon as given in eq (12), (15) and (16), and control inputs are optimized to minimize expected deviations and control effort. In this phase the cost function is given as:

$$U_{online} = \sum_{k=0}^{N_p} (\omega[k]^T Q \omega[k] + u[k]^T R u[k] + \lambda_\omega[k]^T S_\omega \lambda_\omega[k]) \quad (21)$$

Here, N_p indicates the prediction horizon ($k = 0$ to $k = 3$) across which the cost function U_{online} is evaluated and minimized. $\lambda_\omega[k]$ represents the slack variable, which is utilized to lighten the limitations on frequency deviation, allowing for some freedom in the optimization process.

These constraints can be formulated as follow:

$$\omega_{\min} - \lambda_\omega \leq \omega[k] \leq \omega_{\max} + \lambda_\omega, \quad \lambda_\omega \geq 0 \quad (22)$$

$$\begin{bmatrix} f_{\min} \\ V_{\min} \end{bmatrix} - \begin{bmatrix} \lambda_f \\ \lambda_V \end{bmatrix} \leq \begin{bmatrix} \Delta f(k) \\ \Delta V(k) \end{bmatrix} \leq \begin{bmatrix} f_{\max} \\ V_{\max} \end{bmatrix} + \begin{bmatrix} \lambda_f \\ \lambda_V \end{bmatrix}, \quad \lambda_f, \lambda_V \geq 0 \quad (23)$$

A cornerstone of AMPC's online phase is its slack variable mechanism (λ_f, λ_V), which enables dynamic constraint relaxation during extreme grid disturbances. Unlike conventional MPC with rigid deviation bounds, AMPC permits temporary, controlled violations through penalty terms in U_{online} , where the weighting matrix S_ω governs flexibility: lower S_ω values allow greater slack, while higher values enforce strict adherence. For instance, during severe voltage sags, AMPC relaxes λ_V to prioritize reactive power support while maintaining stringent λ_f constraints for frequency stability, aligning with IEEE 1547 fault ride-through requirements. This adaptive constraint handling proves critical in low-inertia grids where traditional control strategies often induce instability through overly restrictive actuation.

The slack variables are initialized at zero for a 50Hz VSM-operated system, ensuring strict constraint adherence under normal conditions. During disturbances, AMPC updates λ_f and λ_V iteratively to balance stability with actuation limits, maintaining frequency within $50 \pm 0.5\text{Hz}$ and voltage within $\pm 5\%$ of nominal values per IEEE 1547.

This proposed AMPC framework uses offline reinforcement learning to minimize the cost function $U_{offline}$ via the COA-jDE algorithm, deriving optimal Q and R under constraints. The online MPC phase then applies these parameters with soft constraints for real-time resilience. This enables our control system to learn from the dynamic grid environment and proactively react to changing operational circumstances.

Furthermore, by introducing soft restrictions via slack variables during the online phase, the AMPC enables a flexible and dynamic response to grid disturbances, which typical MPC approaches frequently fail to address effectively. Although standard MPC focusses solely on reducing deviations in state variables, AMPC reduces the cost function and dynamically changes its control effort depending on real-time grid performance, considerably improving frequency stability and adaptability. This flexibility, together with the optimization based on reinforcement learning, makes AMPC a more robust and effective solution, particularly when considering the integration of renewable energy sources with modern power networks.

Table 3 outlines the key variables utilized in the MPC phase of the Adaptive Model Predictive Control (AMPC) system, with a focus on real-time stabilization using soft constraints on frequency and voltage. The stated parameters provide responsive, adaptive constraint management in order to preserve grid stability.

Fault current control strategies in grid-forming inverters

Grid-Forming Inverters (GFMs) have a serious issue of controlling fault currents due to their limited fault current contribution as compared to traditional synchronous generators. This limitation arises from the necessity to protect its semiconductor components, which creates issues with fault detection and protection coordination. Insufficient fault current can restrict the timely isolation of faults, creating a threat for grid stability and reliability, particularly in low-inertia power systems that are heavily dominated by inverter-based resources.

However, to solve these problems, GFMs adopt the three methods for fault current limitation: virtual impedance, the current-limiting method, and their combination. Virtual Impedance boosts the inverter impedance during faults to limit current and is used for overcurrent protection. Current limiting limits maximum

Parameter	Variable	Values
Prediction Horizon	N_p	3
Control Horizon	N_c	3
Sampling Time	T_s	0.01 s
Slack Variable for Frequency	λ_f	Adaptive
Slack Variable for Voltage	λ_v	Adaptive
Weight on Slack Variables	S	0.1
Maximum Frequency Deviation	$\Delta f_{\max}$	0.5 Hz
Maximum Voltage Deviation	$\Delta V_{\max}$	0.1 pu

Table 3. Parameters for MPC with soft constraints on frequency and voltage in online phase of AMPC control.

Parameter	Variable	Value
Virtual Impedance Resistance Coefficient	R_{vir}	0.1876 pu
Virtual Impedance X/R Ratio	X/R	13.2
Maximum Current Limit	I_{max}	1.2 A
Filter Time Constant	τ_{filter}	1.0 ms
Maximum Saturation Current	$I_{sat,max}$	1.4 pu
Saturation Delay	t_{sat}	1.0 ms
Current Saturation Run Time	t_{run}	1.0 ms

Table 4. Parameters for virtual impedance and current limiting method used in GFM.

output current, thus keeping it below the rated capacity of the inverter. By putting both strategies together, this approach becomes more balanced and it leads towards better fault ride-through capability and improved grid stability in the perspective of controlling fault currents.

In this study, we have utilized the combined Virtual Impedance and Current Limiting technique in all the considered control strategies of GFM (VSM, VSM plus MPC, and AMPC) for a fair comparison. This technique is chosen because it provides improved fault current management and ensures both GFM component protection and microgrid stability during disturbances, making it adequate for testing complex control algorithms. The parameters for the current limiting method, are detailed in Table 4.

The next section details the design and implementation of the COA-jDE algorithm, which is employed during the offline phase to minimize the cost function $U_{offline}$ by optimizing the control parameters Q and R .

Algorithms used in offline reinforcement learning Crayfish Optimization Algorithm (COA)

The Crayfish Optimization Algorithm (COA), introduced by Jia et al. (2023) in³³, is a swarm intelligence algorithm inspired by crayfish behaviors such as foraging, summer resort avoidance, and intraspecies competition. In COA, each crayfish represents a candidate solution, and their positions in the search space are iteratively updated based on environmental factors like temperature, which guides exploration and exploitation.

The algorithm mimics crayfish responses to temperature fluctuations: during “summer resort” phases, crayfish (solutions) retreat to cooler areas (exploitation), while foraging and competitive behaviors drive exploration of new regions. Interactions between individuals balance global search and local refinement, enabling adaptation to complex optimization landscapes. COA’s stages systematically model these behaviors, structuring population dynamics to navigate toward optimal solutions efficiently.

The key steps of the COA are as follows:

Initialization

In multi-dimensional optimization problems, each crayfish represents a solution vector of dimensions $(1 \times \dim)$. Each row in this matrix signifies a solution to the problem. In a given set of variables $(X_{i,1}, X_{i,2}, \dots, X_{i,\dim})$, each variable $X_{i,j}$ must fall within predefined upper and lower boundaries.

The initialization process for the COA involves randomly generating a group of candidate solutions, forming the initial population in the solution space.

These candidate solutions (X) are determined by the population size (N) and the number of dimensions ($\dim$). This random initialization step in the COA algorithm is described by the following equation:

$$X = \begin{bmatrix} X_{1,1} & \cdots & X_{1,j} & \cdots & X_{1,dim} \\ \vdots & \cdots & \vdots & \cdots & \vdots \\ X_{i,1} & \cdots & X_{i,j} & \cdots & X_{i,dim} \\ \vdots & \cdots & \vdots & \cdots & \vdots \\ X_{N,1} & \cdots & X_{N,j} & \cdots & X_{N,dim} \end{bmatrix} \quad (24)$$

In this context, X represents the initial positions of the population. The variable $X_{i,j}$ specifies the position of the i th individual in the j th dimension. The value of $X_{i,j}$ is obtained from eq (25).

$$X_{i,j} = ((u_{b_j} - l_{b_j}) \times rand) + l_{b_j} \quad (25)$$

The j th dimension's lower and upper bounds are represented by l_{b_j} and u_{b_j} , respectively, while a random integer is represented by $rand$.

Temperature parameters and crayfish feeding dynamics

Temperature (t_{temp}) drives behavioral transitions and is modeled as:

$$t_{temp} = (15 \times rand) + 20 \quad (26)$$

where t_{temp} reflects the temperature of the environment in which the crayfish is present.

Crayfish feed optimally at $25^\circ C$, with intake p following a normal distribution (Fig. 3):

$$p = C_1 \left(\exp \left(-\frac{(t - \mu)^2}{2\sigma^2} \right) \times \frac{1}{\sqrt{2 \times \pi \times \sigma}} \right) \quad (27)$$

μ indicates the optimal temperature for crayfish, whereas σ and C_1 regulate crayfish intake at different temperatures.

Summer resort stage (exploration)

When temperatures exceed $30^\circ C$, crayfish migrate to cooler caves. The cave position X_{shade} is computed as the midpoint between the global best (X_G) and current population best (X_L):

$$X_{shade} = \frac{X_L + X_G}{2} \quad (28)$$

If a random number < 0.5 (no competition), crayfish enter the cave (Fig. 4) via:

$$X_{i,j}^{t+1} = C_2 \times rand \times (X_{shade} - X_{i,j}^t) + X_{i,j}^t \quad (29)$$

where C_2 (Eq.(30)) decreases linearly with iterations t , ensuring gradual convergence. This movement toward X_{shade} enhances exploitation of promising regions, accelerating convergence.

$$C_2 = 2 - \frac{t}{T} \quad (30)$$

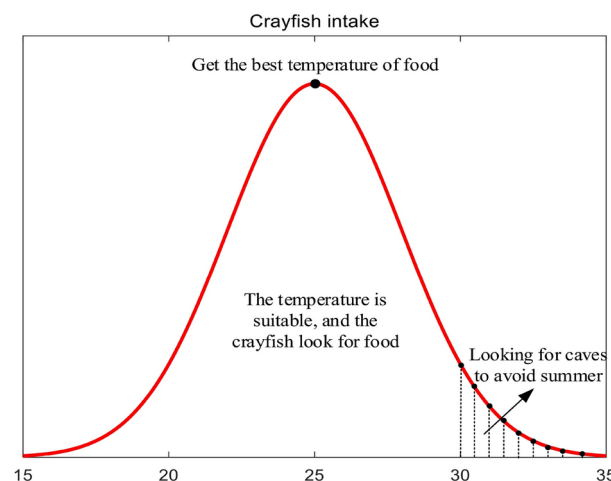


Fig. 3. Influence of temperature on crayfish uptake.

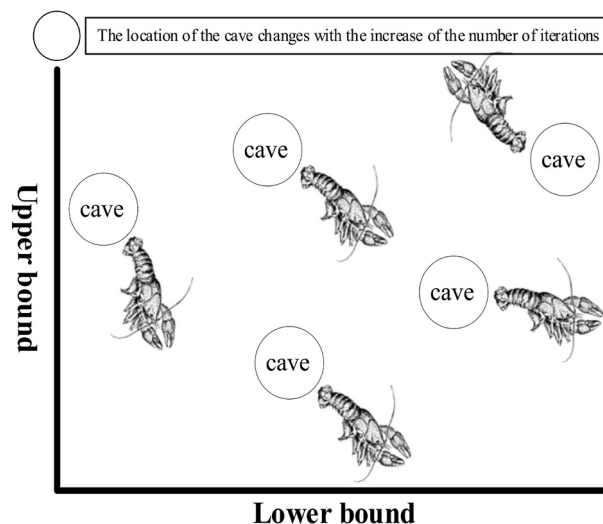


Fig. 4. The Crayfish entering in the cave.

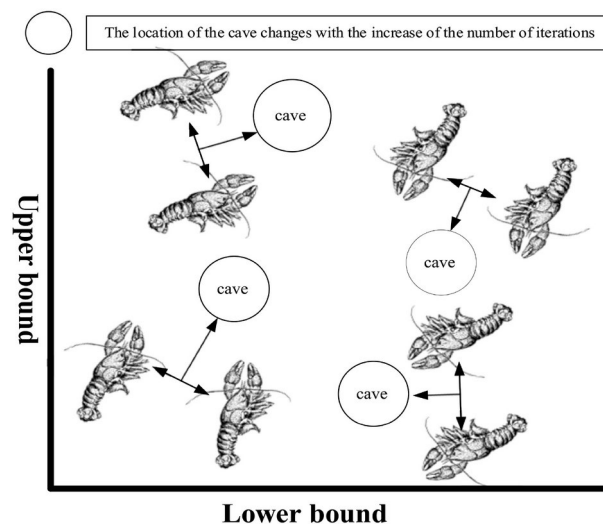


Fig. 5. Crayfish competing to secure the cave.

The Summer Resort stage directs crayfish toward the cave (X_{shade}), which approximates the optimal solution. This movement enhances COA's capability to exploit promising search regions, accelerating convergence and improving efficiency in locating high-quality solutions.

Competition stage (exploitation)

When the temperature exceeds $30^{\circ}C$ and the $rand \geq 0.5$, crayfish compete for cave access (Fig. 5). Positions update via competitive interaction:

$$X_{i,j}^{t+1} = X_{shade} + X_{i,j}^t - X_{z,j}^t \quad (31)$$

where z is a randomly selected crayfish:

$$z = 1 + round(rand \times (N - 1)) \quad (32)$$

This competition introduces perturbations in the population, expanding COA's search scope and enhancing exploration capacity.

Foraging stage (exploitation)

When the temperature is ≤ 30 , crayfish actively seek food. Upon locating a meal, they assess its size. If it is too large, they use their claws to break it down before consuming it with their second and third walking legs. The food source position is given by:

$$X_{food} = X_{G\bullet} \quad (33)$$

The food size, Q , is defined as:

$$Q = C_3 \cdot rand \cdot \frac{f_i}{f_{food}} \quad (34)$$

where $C_3 = 3$ represents the maximum food size, f_i is the fitness of the i^{th} crayfish, and f_{food} is the fitness of the food source.

If $Q > \frac{C_3+1}{2}$, the food is too large and must be broken down, modeled by:

$$X_{G\bullet} = \exp\left(\frac{-1}{Q}\right) \times X_{food\bullet} \quad (35)$$

To simulate the alternating movement of the crayfish's legs while consuming food, a sine-cosine function governs their motion:

$$X_{i,j}^{t+1} = X_{i,j}^t + pX_{food} (\cos(2\pi \cdot rand) - \sin(2\pi \cdot rand)) \quad (36)$$

If $(Q \leq \frac{C_3+1}{2})$, the crayfish move directly towards the food:

$$X_{i,j}^{t+1} = p((X_{i,j}^t - X_{food}) + rand \cdot X_{i,j}^t) \quad (37)$$

The foraging stage enables adaptive movement based on food size, balancing direct exploitation with necessary modifications. This mechanism enhances COA's convergence, improving optimization performance.

The COA algorithm has been rigorously validated on 23 standard test functions and the CEC2014 benchmark suite, demonstrating superior performance in balancing exploration and exploitation for high-dimensional, non-convex optimization tasks in³³. COA's temperature-driven behavioral stages, summer resort avoidance, foraging, and competition, dynamically balance global exploration and local exploitation. This mechanism is particularly effective for offline policy tuning, where computational intensity is permissible, ensuring robust parameter selection (e.g., Q , R) for the AMPC framework while avoiding premature convergence. In this study, COA-jDE's hybrid integration with jDE further refines this process, leveraging adaptive mutation and crossover to minimize the offline cost function $U_{offline}$ (Fig. 2). The convergence curve shows COA-jDE achieving a final cost of 303, enabling AMPC to outperform both MPC and VSM in dynamic scenarios (e.g., 23.6% smaller frequency deviations during load shifts). By unifying offline optimization with online MPC, our framework retains MPC's advantages (adaptive control, constraint adherence, and predictive power) while achieving real-time deployability in low-inertia grids.

Differential Evolution Optimization Algorithm (DE)

Differential Evolution (DE) is a robust and efficient evolutionary optimization algorithm, developed by Storn and Price in 1997, that is widely used across various optimization problems. DE is population-based, relying on iterative mechanisms such as mutation, crossover, and selection to evolve solutions. It efficiently navigates large and complex search spaces, making it popular in optimization tasks³⁴.

While DE and genetic algorithms (GA) share similarities in using population-based methods, DE distinguishes itself by emphasizing mutation over crossover for solution improvement. Mutation, based on differences between randomly selected population members, helps explore the search space effectively.

The fundamental procedure of the DE algorithm after the initialization of the algorithm is outlined as follows:

Mutation operation

DE generates trial vectors by perturbing the population. For each individual x_r^t , a mutant vector v_i^{t+1} is created by adding the weighted difference between two randomly chosen population vectors:

$$v_i^{t+1} = x_r^t + F \cdot (x_{r1}^t - x_{r2}^t) \quad (38)$$

where F is the mutation factor, typically between 0 and 2, and x_{r1}^t , x_{r2}^t are distinct random population members.

Crossover operation

After mutation, DE uses crossover to combine the mutant vector with the target vector. The trial solution u_i^{t+1} is formed as follows:

$$u_i^{t+1} = \begin{cases} v_i^{t+1} & \text{if } rand \leq CR \\ x_i^t & \text{if } rand > CR \end{cases} \quad (39)$$

where CR is the crossover rate, and $rand$ is a random number in the range $(0, 1)$.

Selection mechanism

After mutation and crossover, the trial vector u_i^{t+1} is compared to the target vector x_i^t . If u_i^{t+1} is fitter, it replaces x_i^t in the population:

$$x_i^{t+1} = \begin{cases} u_i^{t+1} & \text{if } U(u_i^{t+1}) < U(x_i^t) \\ x_i^t & \text{otherwise} \end{cases} \quad (40)$$

Here, U represents the fitness function to minimize.

This selection process ensures that the population evolves towards better solutions over successive generations.

Self-Adaptive Differential Evolution (jDE)

Traditional Differential Evolution (DE) uses fixed values for the mutation factor F and crossover probability CR throughout the process. However, because the optimal settings for these parameters depend on the problem's complexity,³⁵ introduced Self-Adaptive Differential Evolution (jDE), which adjusts F and CR dynamically. This adaptive mechanism tailors the search strategy to the problem at hand, improving efficiency, especially for high-dimensional and complex optimization tasks. In jDE, the control parameters are updated as follows:

$$F_i^{t+1} = \begin{cases} F_l + rand_1 \times F_u & \text{if } rand_2 < \tau_1 \\ F_i^t & \text{otherwise} \end{cases} \quad (41)$$

$$CR_i^{t+1} = \begin{cases} rand_3 & \text{if } rand_4 < \tau_2 \\ CR_i^t & \text{otherwise} \end{cases} \quad (42)$$

Here, τ_1 and τ_2 are conversion probabilities (both set to 0.1), and $rand$ represents a random number. F_l and F_u denote the lower and upper bounds for F , which is initially set to 0.5, while CR starts at 0.9. These equations ensure that each search agent maintains its own F and CR , with occasional updates due to the low transition probability.

Hybrid COA-jDE for cost function minimization in offline reinforcement learning

The integration of COA and jDE addresses a fundamental challenge in metaheuristic optimization: balancing global exploration and local exploitation across diverse solution landscapes. This hybrid approach is not merely a sequential combination as seen in^{36–39} but a dynamic, fitness-driven interaction where COA and jDE alternately dominate based on population fitness, ensuring sustained optimization momentum.

The integration of COA-jDE into the AMPC framework is motivated by its proven efficacy in navigating high-dimensional, non-convex optimization landscapes, essential for tuning the AMPC's weight matrices Q and R . Unlike traditional reinforcement learning methods that depend on stochastic gradient updates, COA-jDE synergizes global exploration (via COA's temperature-driven summer resort/foraging stages) with local exploitation (through jDE's self-adaptive mutation/crossover). This hybrid strategy enables broad parameter space sampling to evade local minima while refining solutions for precise convergence. As evidenced in Fig. 2, COA-jDE achieves a 95% reduction in $U_{offline}$ within 3,000 iterations, demonstrating superior capability in balancing exploration-exploitation tradeoffs. By decoupling offline learning from online execution, this approach preserves MPC's real-time adaptability while circumventing the computational overhead and instability risks inherent to online reinforcement learning.

Below, we explain the technical rationale and empirical advantages underpinning this synergy.

1) Adaptive algorithm switching: COA-jDE employs a fitness-dependent switching mechanism:

- a) **Below-average fitness individuals** activate jDE for localized exploitation. jDE's self-adaptive mutation factor F and crossover rate CR (Eqs.(41)-(42)) refine solutions through stochastic differential mutation, enhancing convergence in hybrid frameworks^{40–42}.
- b) **Above-average fitness individuals** trigger COA's temperature-driven exploration (summer resort/foraging stages). COA's biologically inspired dispersion mimics multi-swarm algorithms, targeting distinct search regions^{33,43–45}.

This conditional switching prevents resource wastage on unpromising solutions while aggressively exploring high-potential regions.

- 2) **Complementary search dynamics:** COA's deterministic temperature transitions (Eq.(26)) provide structured exploration but lack granularity in exploitation. jDE compensates through stochastic differential mutation (Eq.(38)), injecting controlled randomness into high-fitness solutions^{46,47}. This "explorative exploitation" aligns with self-adaptive DE variants like SHADE, adaptively scaling perturbations to maintain diversity⁴⁸.
- 3) **Mitigating parameter sensitivity:** Standalone COA requires manual tuning of food intake (C_1) and competition parameters, which can degrade performance in dynamic environments due to rigid parameter dependencies⁴⁵. jDE's self-adaptation automates parameter control via zoning evolution⁴⁶, partitioning optimization into phases with distinct adaptation rules, significantly reducing sensitivity to initial settings.

This synergy translates into five key advancements over standalone COA: (1) automated parameter adaptation, (2) accelerated convergence, (3) escape from deceptive local optima, (4) robustness to initial population bias, and (5) superior solution precision. By replacing COA's static parameters with jDE's self-adaptive mutation (F) and crossover (CR) via zoning evolution⁴⁶, COA-jDE eliminates manual tuning dependencies, ensuring consistent performance across dynamic environments critical for offline RL. jDE's stochastic mutation (Eq(38)) refines solutions more efficiently than COA's deterministic foraging, accelerating convergence in complex landscapes like high-dimensional control tasks³³. Furthermore, jDE's success-history-based mutation⁴⁸ dynamically scales perturbations to escape local basins where COA stagnates, while adaptive mutation counters initial population bias, maintaining solution consistency⁴⁵. Finally, COA-jDE synergizes COA's global exploration with jDE's granular exploitation, outperforming standalone COA in precision-critical applications like robotics trajectory optimization. These enhancements establish COA-jDE as a robust optimizer for offline RL, where balancing exploration and exploitation is pivotal to minimizing $U_{offline}$ and securing globally optimal policies.

Fig. 6 presents the flowchart for reinforcement learning based on the COA-jDE algorithm, with its parameters detailed in Table 5.

The procedural steps for implementing the COA-jDE hybrid algorithm are detailed below:

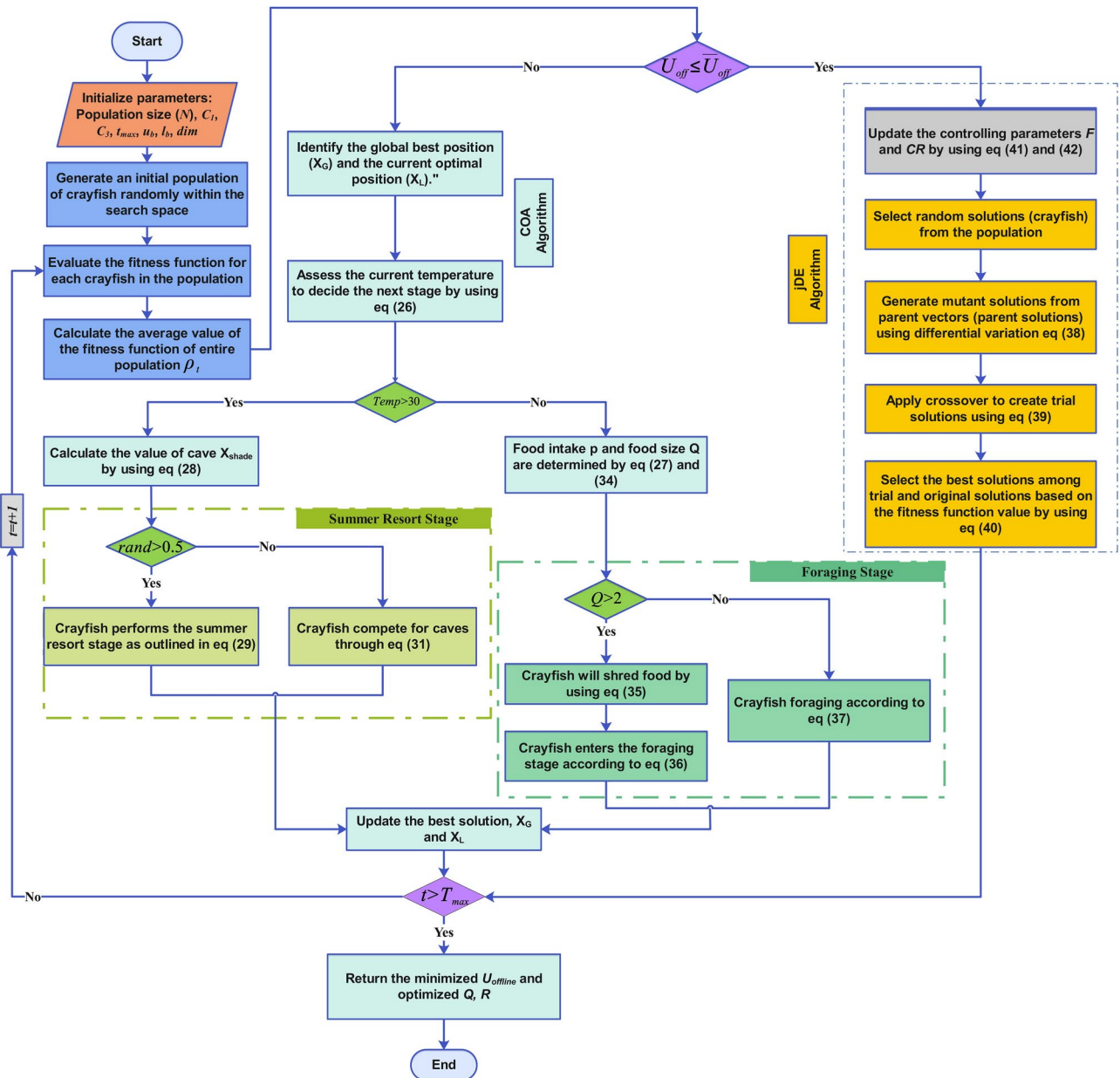


Fig. 6. Implementation of the COA-jDE algorithm for cost function minimization ($U_{offline}$) in offline reinforcement learning for AMPC control.

Algorithm	Parameter	Variable	Value
Crayfish Optimization Algorithm	Influences crayfish individual movement	C_1	0.4
	Related to positioning adjustments	C_3	3.5
	Scaling factor for search scope	μ	31
	Inertia weight affecting movement influence	σ	3
	Population size	N	1000
	Lower and upper bounds of search space	l_b, u_b	[-10, 10]
	Dimensionality of search space	dim	4
Self-Adaptive Differential Evolution Algorithm	Scaling factor for differential mutation	F	0.5
	Crossover rate	CR	0.9
	Mutation control parameter adjustment threshold	T_F	0.1
	Crossover control parameter adjustment threshold	T_{CR}	0.1

Table 5. COA-jDE optimization algorithm parameters for Offline reinforcement learning of ρ_t .

- 1) **Initialize parameters:** Define population size (N), search bounds (u_b, l_b), dimensions (dim), and maximum iterations (t_{max}). These parameters govern the algorithm’s exploration-exploitation behavior for minimizing $U_{offline}$.
- 2) **Generate initial population:** Create a random population within the search space to ensure diversity, enabling exploration of solutions that minimize $U_{offline}$.
- 3) **Evaluate fitness function:** Calculate each individual’s fitness as $U_{offline}$. Lower values indicate better performance (reduced frequency deviations/control effort).
- 4) **Calculate average fitness ($\bar{U}_{offline}$):** The average fitness value for the entire population is determined. This serves as a benchmark for determining whether to use the jDE or COA algorithm for each iteration, hence guiding the optimization strategy.
- 5) **Conditional selection of algorithm:**

The algorithm prioritizes exploitation via jDE for solutions where $U_{offline}$ is below or equal to the population average ($\bar{U}_{offline}$), refining these candidates through mutation and crossover. For solutions with higher $U_{offline}$, it triggers exploration using COA, enabling global search to escape local minima. This adaptive switching ensures efficient resource allocation: refining promising solutions while aggressively exploring underperforming regions, accelerating convergence to the global optimum.

- 6) **Crayfish optimization (COA) stages:**
 - a) **Summer resort stage:** During this stage, crayfish relocate to more desirable locations based on simulated environmental circumstances, such as temperature changes. The position of each crayfish X_i is updated using equation (29). This phase facilitates large-scale exploration of the search space by modifying the crayfish positions based on the best available solutions and ambient randomization, so preventing premature convergence.
 - b) **Foraging stage:** In this stage, crayfish compete for caves to find better food sources. Equation (36) models the behavior. This step refines the exploration started in the summer resort stage by focusing on certain high-potential places in the search space. It allows the population to focus on the best solutions while retaining some randomness, preventing the algorithm from becoming trapped in local optima.
- 7) **jDE operations:**
 - a) **Mutation and crossover:** Solutions are modified using differential variation and crossover procedures. These phases generate novel outcomes by combining features from parent solutions, increasing diversity while refining the search.
 - b) **Selection:** The most effective responses from the trial and original solutions are selected based on their fitness values. This retains solutions with lower $U_{offline}$. Which ensures that only the fittest solutions are maintained.
- 8) **Update the controlling parameters (F and CR):** The COA and jDE dynamically update their regulating parameters (e.g., food intake, mutation rate F , and crossover rate CR) as they advance. This adaptability guarantees that the method remains effective during various stages of optimization.
- 9) **Update the global best solutions (X_G and X_L):** X_G tracks the global minimum of $U_{offline}$ across all iterations, preserving the best Q and R . X_L records the current iteration’s minimum, guiding immediate refinement while ensuring adaptability to new solutions. These positions represent the most effective solutions discovered thus far and direct the population’s progress toward the global optimum.
- 10) **Return maximized reward with optimized variables:** When the maximum iteration or stopping condition is met, the method returns the minimized $U_{offline}$ and optimized Q, R . These parameters will subsequently be used in the online control phase to ensure the best system performance.

Terminate optimization if $U_{offline}$ shows no significant decrease after 50 – 100 iterations (Section 3.1), ensuring thorough search-space exploration.

The COA-jDE algorithm minimizes $U_{offline}$ by strategically alternating between COA (exploration) and jDE (exploitation). This hybrid approach converges to optimal Q and R values, ensuring robust AMPC performance in dynamic grids.

The following section presents the experimental findings from the microgrid setup, illustrating the effectiveness and precision of the suggested AMPC control technique.

System Configuration

The microgrid (MG) under consideration is a 16 MW grid-connected system powered by eight 2 MW DFIG wind turbines. These turbines connect to a 25 kV distribution bus (B5) via a 25 MVA three-phase transformer, ensuring effective power transfer from the wind farm to the network.

A considerable load is connected to Bus B3, which consists of four 3 MVA units operating at 480 V. These loads include an induction motor equivalent to 2.73 MW at a power factor of 0.91 and a 2 MVA induction motor at a power factor of 0.9, which constitutes 1.82 MW to the system.

Furthermore, the local grid communicates with the microgrid via Bus B2, which is connected by a 25 MVA transformer. This connection allows the microgrid to maintain synchronization with the main power grid.

The complete test setup is modeled and simulated in MATLAB/Simulink, and each case study is thoroughly analyzed. The simulation employs a time-domain solver with a step size of $100\mu s$ to accurately analyze the system's dynamic performance.

The primary goal of this study is to assess the efficacy of the suggested Adaptive Model Predictive Control (AMPC) method when applied to the GFM. The results will be compared to conventional VSM control and VSM with MPC under various operating situations, while closely adhering to the grid codes listed in Table 1.

Fig. 7 shows the system's single-line diagram. Table 6 contains detailed specifications for the GFM, and Table 7 describes the wind turbines and DFIG generators.

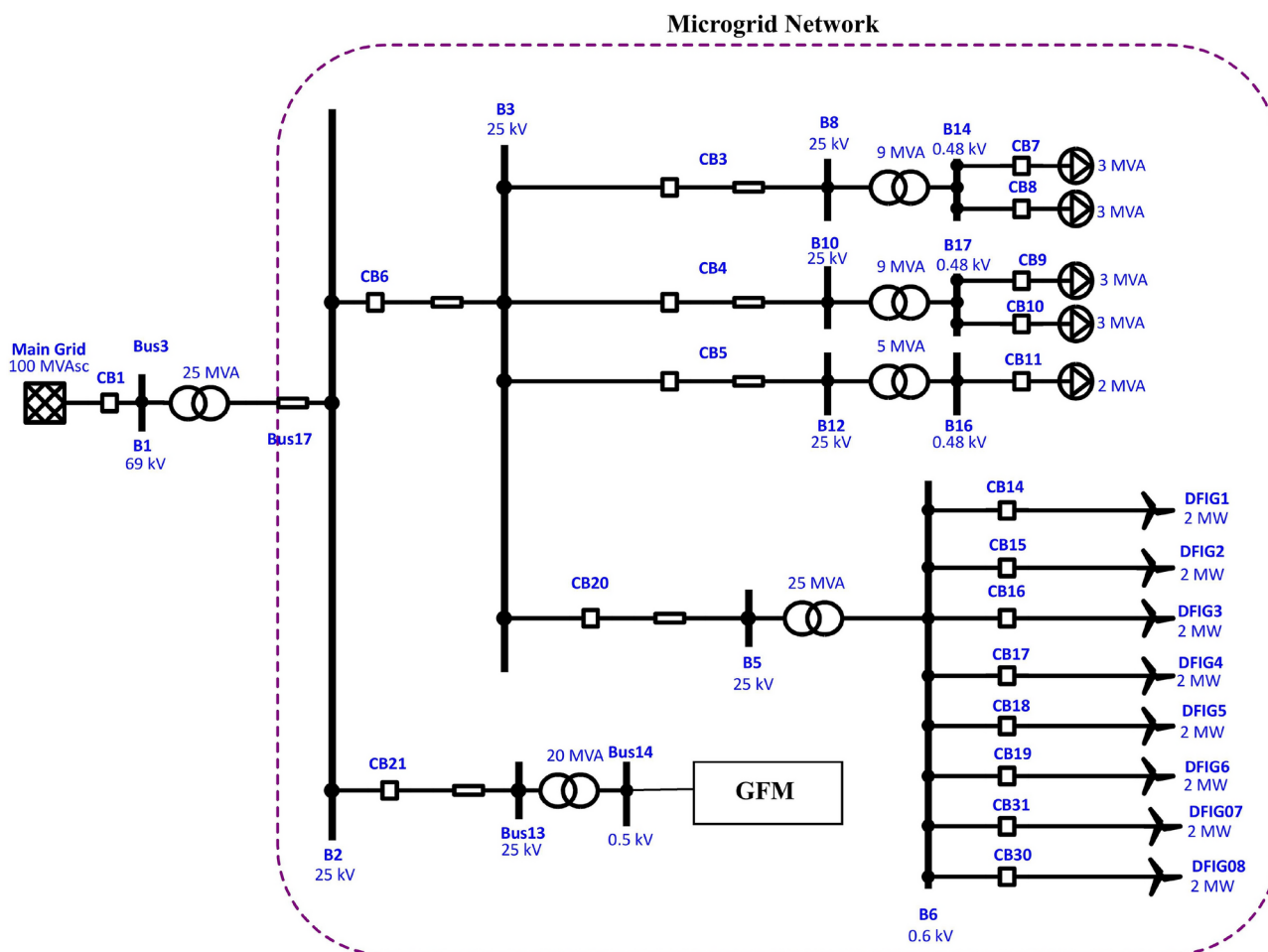


Fig. 7. Single-line diagram of the 16 MW grid-connected microgrid with DFIG wind turbines and GFM.

Parameter	Variable	Value
Apparent Power	S	16.8 MVA
Grid Frequency	f	50 Hz
DC Bus Voltage	V_{DC}	1500 V
Line RMS Voltage	V_L	0.5 kV
Measurement Sample Time	T_{sample}	100 μs
Grid X/R Ratio	X/R	4
Short-Circuit Ratio (SCR)	SCR	2.5
Inverter Efficiency	η	97 %

Table 6. Key parameters for Grid-Forming inverter (GFM).

Parameters	Values
Rating, Voltage, Frequency	16MW (8 unit), 0.6kV, 50Hz
Rotor/Stator Inductance	$L_r = 0.157\text{ H}$, $L_s = 0.172\text{ H}$
Rotor/Stator Resistance	$R_r = 0.005\text{ }\Omega$, $R_s = 0.00705\text{ }\Omega$
Nominal wind speed	12m/s
Cut-in wind speed	3.5m/s
Cut-out wind speed	25m/s
No. of blades	3
Diameter of Rotor	82.5
Area swept by the blades	5343m ²
Rotor speed	10.1 – 18.7rpm
Max allowed pitch angle rate	2deg/s

Table 7. Parameters of DFIGs in the microgrid.

Results and discussion

To identify the optimal place for GFM in the microgrid (MG) configuration, independent tests were conducted on various buses, with an emphasis on bus B2 (PCC) and bus B3. The MG has a peak load of 14MVA and the total generation capacity from 8 DFIG wind turbines is 16MW. The GFM’s performance was assessed based on its capacity to deliver both active and reactive power to stabilize frequency and voltage under a variety of conditions, including grid disturbances, faults, and voltage fluctuations. After conducting a detailed study, it was found that optimum placement of GFM at bus B2 offered best results for overall system stabilization. As PCC is the major interface between the MG and local grid (interfaced by a 69/25kV transformer), it enabled the GFM to more efficiently regulate power exchange, resulting in improved voltage regulation and frequency stability. This placement also ensured effective power flow management during transitions between grid-connected and islanded modes, as well as quick response to disruptions, ensuring the microgrid’s overall stability.

To determine the size of GFM, we took into account the peak load connected to the MG (14MVA) and the requirement for appropriate active and reactive power support for frequency and voltage stability in this MG. The GFM is designed to handle peak loads with a 20 – 30% margin, guaranteeing that it could handle fluctuation in load and generation under both normal and severe fault conditions. This margin enables the GFM to deliver adequate synthetic inertia and reactive power during important events such as 3-phase faults, voltage sags, or load shifts. The GFM, interfaced with a transformer of 0.5/25kV to the PCC, ensured that the microgrid could operate reliably even when disconnected from the local grid. By matching the GFM’s size to the system’s dynamic requirements, we assured that it could effectively manage power flow, thus increasing the microgrid’s resilience and stability under a wide range of operational situations.

In this section, the GFM’s performance is thoroughly analyzed across various operating conditions to determine the effectiveness of the suggested AMPC technique. The test cases chosen include changes in local load reference, abrupt grid frequency reference shift, islanding, and shifts in grid voltage phase angle. These cases were chosen because they exhibit severe disturbances that directly affect network frequency, which is crucial to grid stability. These perturbations show the AMPC’s frequency stabilization durability under realistic situations.

In addition, an extreme three-phase (LLG) fault is simulated to evaluate voltage ride-through capabilities. This scenario brings severe voltage disruptions that might result in major dips or surges, endangering the operation of the microgrid network. The designed AMPC must be tested against such failures to assure system stability under worst-case voltage circumstances.

This thorough investigation is essential to validating the robustness of the AMPC when compared to conventional MPC and traditional VSM strategies, so proving the superiority of the proposed method in preserving grid stability and performance during dynamic disturbances in modern power systems. The parameters of the VSM control and MPC control employed in the comparative study against AMPC are given in Tables 8 and 9.

Parameter	Variable	Value
Inertia Constant	H	1 s
Damping Coefficient	D	1.056 pu/Hz
Frequency Droop	K_f	10 pu
Power Measurement Filter Time Constant	τ_p	1.00e-03 s
Maximum Damping Power	$P_{d,max}$	0.7 pu
Minimum Damping Power	$P_{d,min}$	-0.6 pu
Sampling Time	T_s	0.01 s

Table 8. Key parameters for virtual synchronous machine (VSM) control mode in grid-forming inverter.

Parameter	Variable	Value
Prediction Horizon	N_p	5 steps
Control Horizon	N_c	3 steps
Sampling Time	T_s	0.01 s
Weight on Frequency Error	w_f	1
Weight on Voltage Error	w_v	0.5
Weight on Control Effort	w_u	0.01
Inertia Weight	J	1.5

Table 9. Parameters for Traditional Model Predictive Control (MPC) in VSM mode for grid-forming inverter with emphasis on frequency stability.

The following scenarios are used to assess the performance of the AMPC-controlled GFM in a microgrid environment:

- 1) GFM Synchronization with Local Grid Reference Frequency Shifts
- 2) GFM Frequency Stability Response to Local Load Variations
- 3) Impact of Grid Voltage Phase Angle Variations on PCC frequency
- 4) Performance of GFM During Islanding Scenarios
- 5) Assessment of GFM’s LVRT Performance Under Symmetrical Grid Faults

GFM synchronization with local grid reference frequency shifts

In this case, the response of the GFM is studied when the local grid frequency, taken as the reference in grid-connected mode, changes. As outlined in GC0137 and mentioned in Table 2, GFMs must accurately track and follow local grid frequency variations, maintaining synchronization with the grid. This involves adjusting their output in real-time to align with frequency deviations, ensuring system stability by providing dynamic responses to both frequency increases and decreases.

To evaluate the GFM’s synchronization capability and rapid response to changes in the local grid frequency, two scenarios are considered. First, the reference frequency is increased from 50Hz to 52Hz at a rate of 2Hz/s and held at 52Hz for 5 seconds. Then, the reference frequency is decreased from 52Hz to 47Hz at a rate of 1Hz/s. These scenarios allow for testing of both increasing and decreasing frequencies, as well as different frequency change rates. This thorough analysis assesses the tracking performance of all three control strategies: AMPC, MPC, and VSM.

In Fig. 8, at $t = 4s$, the grid reference frequency (yellow line) is adjusted from 50Hz to 52Hz. In the zoomed window, it is evident that AMPC, shown in green, responds faster than the other controls. This is attributed to its advanced predictive capability, which allows it to anticipate future states. AMPC begins tracking the frequency change at 4.01s, whereas MPC responds at 4.06s, and VSM follows at 4.12s. Furthermore, at $t = 5s$, the AMPC exhibits less overshoot compared to MPC, while MPC outperforms VSM in this regard. Similar fast-tracking responses from AMPC are observed at $t = 10s$ and $t = 15s$ when the frequency decreases from 52Hz to 47Hz, demonstrating the superior predictive performance of AMPC over MPC.

In Fig. 9, the active power output of the GFM during frequency synchronization is illustrated. At $t = 4s$, as the local grid frequency increases to 52Hz, AMPC immediately starts absorbing active power from the PCC. This response occurs because an increase in grid frequency typically indicates that generation exceeds demand, creating an excess of power in the system. To maintain grid stability and avoid frequency overshoot, the GFM absorbs this excess power, helping to restore the balance between supply and demand. As the frequency settles at 52Hz, the AMPC control shows less overshoot compared to other methods, reflecting its ability to respond more smoothly to changes. At all transition points, the AMPC demonstrates superior efficiency in tracking the frequency and synchronizing the GFM output with the grid compared to the other control strategies.

According to GC0137, as outlined in Table 2, the GFM must follow the reference frequency accurately in grid-connected mode, maintaining a stable connection to the local grid. To test the robustness of the control strategies under extreme conditions, the frequency is held at both 52Hz and 47Hz for at least 5 seconds to assess

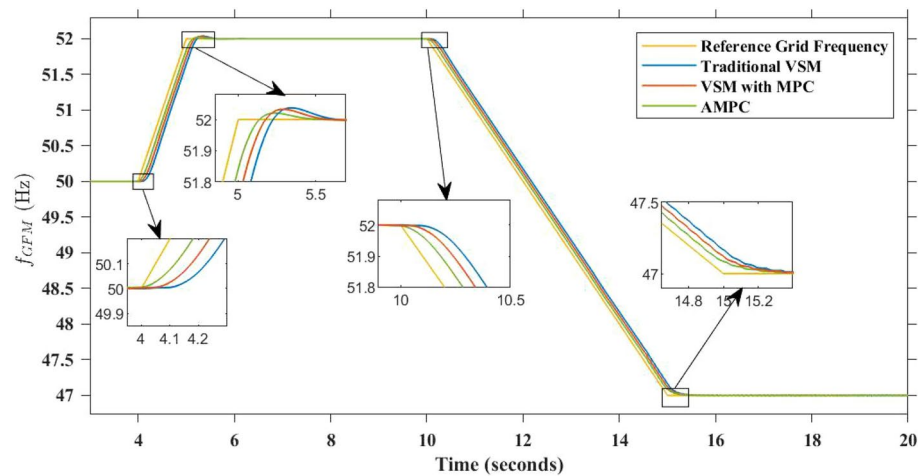


Fig. 8. GFM frequency response to sudden grid frequency variations from 50Hz to 52Hz at $t = 4s$, and a drop to 47Hz at $t = 10s$.

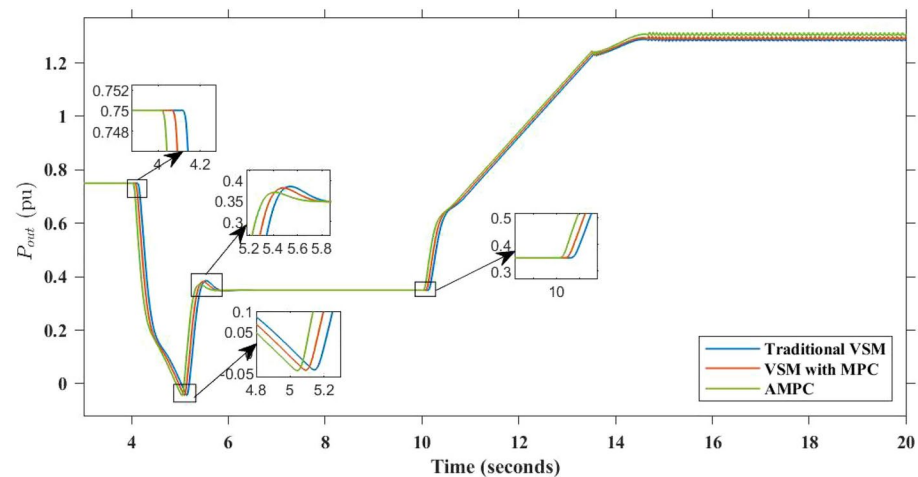


Fig. 9. Active power output from GFM to track local grid frequency changes.

the low- and high-frequency ride-through capability of the GFM. From this analysis of frequency variations, it is clear that the proposed AMPC provides a significantly faster response and superior synchronization compared to traditional MPC and VSM controls.

GFM frequency stability response to local load variations

In this case, the impact of sudden microgrid load variations on the PCC frequency is analyzed for all control schemes: Virtual Synchronous Machine (VSM), conventional MPC, and the proposed AMPC. For easier comparison, all variables are presented in per unit (pu). Initially, the total microgrid load is $0.45pu$. At $t = 4s$, the load is abruptly increased to $0.65pu$, held for 2 seconds, and then reduced to $0.35pu$ at $t = 6s$. These two events, sudden load increase and decrease, are examined to assess their effects on the PCC frequency across the three control strategies.

Fig. 10 illustrates the load power variations for all three control systems. At $t = 4s$, the load increase causes a slight overshoot in all control schemes. AMPC (green line) shows the least overshoot at $0.68pu$, followed by MPC at $0.69pu$, and VSM at $0.694pu$. Additionally, AMPC reaches the steady-state value of $0.65pu$ more quickly than the other two, demonstrating faster settling. During the load decrease at $t = 6s$, an undershoot occurs in all control schemes. Once again, AMPC exhibits smoother recovery and less undershoot, settling faster than both MPC and VSM. This clearly highlights the improved performance of AMPC in mitigating load disturbances.

Fig. 11 shows the frequency response at the PCC during load changes. At $t = 4s$, when the load is increased, the frequency undershoot for AMPC is $49.96Hz$, while MPC and VSM experience larger drops to $49.94Hz$ and $49.92Hz$, respectively. AMPC not only reduces the undershoot but also stabilizes the frequency more quickly. At $t = 6s$, as the load is decreased, the frequency overshoot for AMPC is $50.05Hz$, which is lower than MPC's

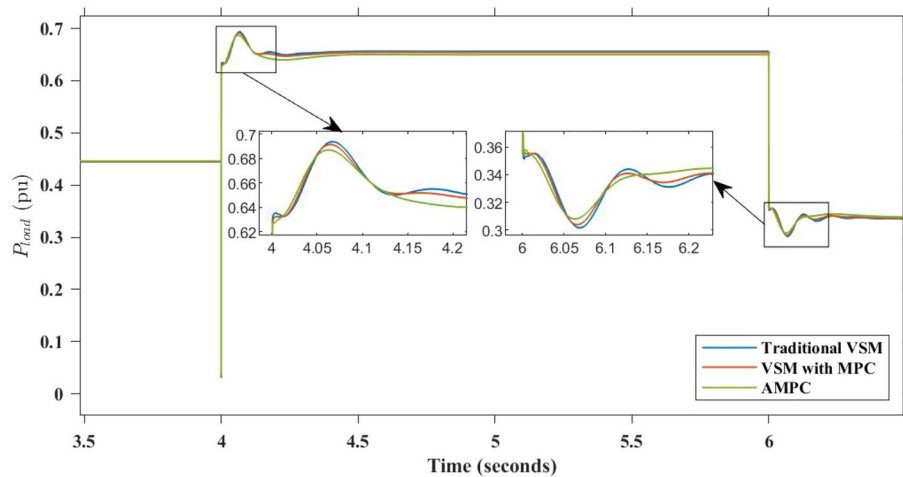


Fig. 10. Step Change in Load Power from $0.45pu$ to $0.65pu$ at $t = 4s$ and from $0.65pu$ to $0.35pu$ at $t = 6s$.

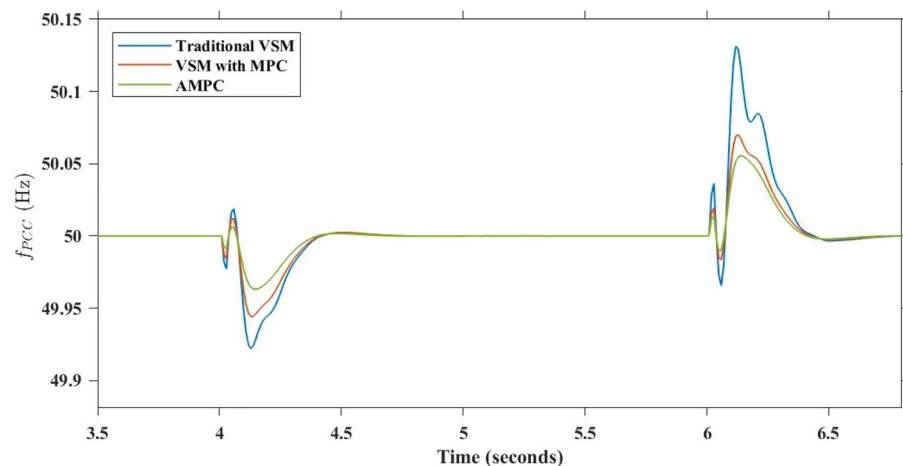


Fig. 11. Frequency Response at PCC during Load Power Shifts from $0.45pu$ to $0.65pu$ and $0.65pu$ to $0.35pu$.

50.07Hz and VSM's 50.13Hz. These results clearly indicate that AMPC minimizes frequency deviations more effectively, leading to better frequency stability during dynamic load changes.

Fig. 12 presents the GFM's power response to the frequency fluctuations at the PCC. When the load increases at $t = 4s$, the GFM injects active power to counteract the frequency dip. AMPC injects $0.79pu$, MPC injects $0.78pu$, and VSM injects $0.76pu$. As the load decreases at $t = 6s$, the GFM absorbs power to mitigate the frequency rise, with AMPC going up to $0.05 pu$, MPC going to $0.07pu$, and VSM going down to $0.10pu$. A small overshoot occurs around $t = 6.2s$, caused by the GFM's action to stabilize the frequency after the load reduction.

IEEE 1457 criteria state, as indicated in Table 2, that the GFM must offer continuous frequency support within a $\pm 2\%$ tolerance range. This condition is met in all control situations, including conventional MPC, VSM, and the proposed AMPC. Table 10 summarizes the frequency fluctuations at the PCC during load variations.

These graphical findings clearly indicate the effectiveness of the suggested AMPC in boosting the GFM's virtual inertia capabilities, particularly during unexpected load fluctuations, while also greatly improving stability when compared to existing VSM and MPC controls.

Impact of grid voltage phase angle variations on PCC frequency

In this case, the impact of a phase shift in the local grid voltage on the PCC frequency is analyzed to evaluate the synchronization and stability capabilities of the GFM. The primary goal is to assess how effectively the GFM maintains frequency stability and delivers the appropriate power response when subjected to sudden changes in the grid voltage phase angle. By doing so, it demonstrates the GFM's capacity to handle phase angle variations while ensuring stable and reliable grid operation in grid-connected mode.

A 10-degree phase shift is applied at $t = 4s$ in the grid voltage phase angle. Fig. 13 illustrates the resulting frequency fluctuation at the PCC. Initially, a spike in frequency is observed, followed by a slight dip and a second, more significant spike. Notably, the developed AMPC control exhibits the smallest spike, peaking at 50.06Hz,

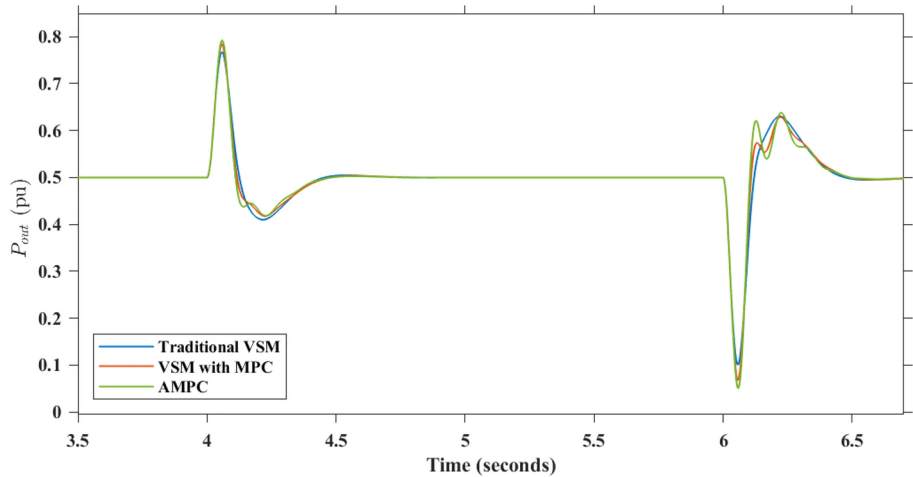


Fig. 12. GFM output power response to load variation for frequency stabilization at PCC.

Case	Time	Traditional VSM	MPC	AMPC
Load Increase US_{max}	4s	49.92 Hz	49.94 Hz	49.96 Hz
Load decrease OS_{max}	6s	50.13 Hz	50.07 Hz	50.05 Hz

Table 10. Numerical summary of frequency fluctuations at the PCC during load variations.

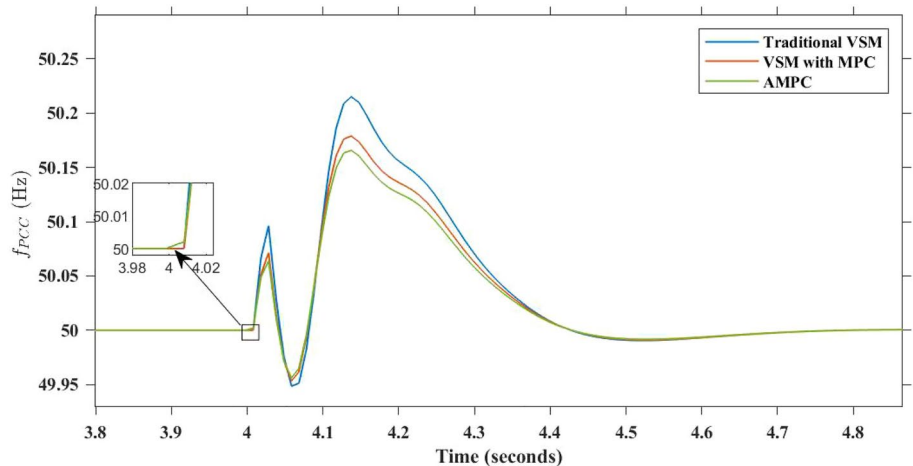


Fig. 13. Frequency Response at PCC During a 10-Degree Phase Shift in Local Grid Voltage at $t = 4s$.

while the MPC and VSM reach 50.07Hz and 50.1Hz, respectively. Afterward, AMPC also shows a smaller dip compared to the other two control strategies. For the second spike, AMPC reaches a maximum overshoot of 50.16Hz, which is lower than 50.18Hz for MPC and 50.22Hz for VSM, before eventually stabilizing back to 50Hz. These results clearly demonstrate that AMPC produces the least deviation, underscoring its superior performance in minimizing frequency disturbances during phase shifts.

Fig. 14 shows the GFM's active power response to this abrupt grid voltage phase shift across the different control methods. The GFM initially absorbs power to manage the frequency rise at the PCC, followed by injecting active power to counterbalance the subsequent dip. The most significant observation is that the AMPC-controlled GFM responds faster than both MPC and VSM, as evidenced by the green line in the graph, which dips earlier and more sharply. This quick and precise reaction is due to the more advanced predictive learning strategy of AMPC, allowing it to anticipate grid changes more efficiently than traditional methods. A summary of frequency deviations at the PCC resulting from grid phase angle shifts is shown in Table 11.

These graphical results clearly highlight the effectiveness and significant improvements in the VSM control of the GFM using the proposed AMPC. The AMPC's adaptive, model-predictive nature allows it to foresee and

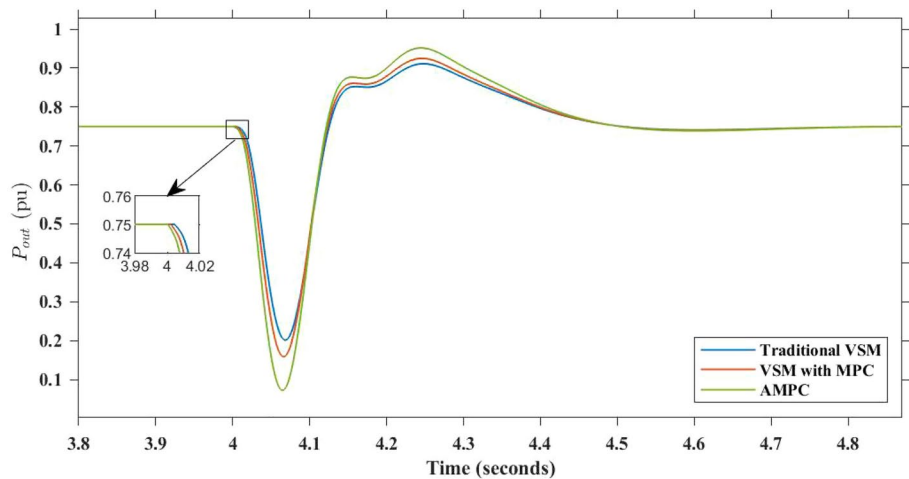


Fig. 14. GFM Active Power Output in Response to a 10-Degree Phase Shift in Local Grid Voltage at $t = 4s$.

Case	Time	Traditional VSM	MPC	AMPC
Phase shift OS_{max}	4s	50.22 Hz	50.18 Hz	50.16 Hz

Table 11. Numerical summary of frequency deviations at the PCC during grid phase angle shifts.

react to disturbances more intelligently, reducing frequency spikes and dips while maintaining system stability. The proposed AMPC proves to be a highly efficient solution for ensuring rapid, accurate, and stable performance during dynamic grid conditions, particularly under phase angle variations.

Performance of GFM during islanding scenarios

The islanding case represents a crucial scenario for evaluating the stability and reliability of GFM, as it directly tests their capability to maintain grid stability when disconnected from the main grid. According to GC0137 standards, GFMs must operate autonomously in islanded mode, demonstrating their ability to uphold frequency and voltage stability within specified limits without relying on external grid support. This functionality is essential for ensuring an uninterrupted power supply and maintaining microgrid system stability during grid separation events. In this study, an islanding event occurring at $t = 4s$ is utilized to assess the performance of the GFM, specifically its capacity to stabilize frequency and balance power during this operation. Such tests are vital for confirming the GFM’s low-inertia performance and its ability to sustain grid synchronization and support under these challenging conditions.

Fig. 15 illustrates the frequency behavior at the PCC during a sudden islanding event at $t = 4s$. All control methods exhibit a minor overshoot like 50.014Hz for AMPC, 50.019Hz for MPC, and 50.027Hz for VSM. After the initial overshoot, the frequency gradually declines in each case. It eventually settles at 49.89Hz for AMPC, 49.8Hz for MPC, and 49.71Hz for VSM. Although all steady-state frequencies remain within the GC0137 std limits, AMPC demonstrates a significant performance improvement over traditional MPC. This is because AMPC’s strong predictive capabilities enable it to better predict system changes and alter its response in real time. Consequently, AMPC reduces both the overshoot and the settling time, ensuring better frequency stability during events like islanding. It maintains system balance better than MPC and traditional VSM by precisely adjusting the frequency to the reference value.

Fig. 16 depicts the GFM’s active power response across all controls. As the microgrid enters standalone mode, the GFM boosts its power injection from 0.6pu to around 0.85pu to support the system. Particularly, the AMPC (green line) has the fastest response, rapidly increasing power output to stabilize the system. This highlights AMPC’s stronger predictive abilities, which enable it to respond more quickly to grid disturbances. The MPC (orange line) responds quickly as well, but it lags slightly behind the AMPC in terms of time and magnitude. In contrast, the typical VSM (blue line) has a delayed reaction, indicating a slower adaptability to rapid changes.

Following the initial response, all scenarios experience a small overshoot at $t = 4.04s$, which is caused by the system’s efforts to balance power and frequency. However, AMPC’s overshoot is far lower, indicating improved damping and control efficacy. MPC and VSM have bigger overshoots due to their slower response dynamics. Furthermore, AMPC’s power output stabilizes faster than the other approaches, demonstrating its efficacy in restoring system stability following an islanding event.

The faster rise time, lower overshoot, and faster settling noticed in AMPC demonstrate its greater capacity to manage dynamic grid changes. This improved performance can be attributed to AMPC’s advanced offline training using the COA-jDE algorithm, which allows for accurate real-time corrections during grid disruptions, as shown in the response curves. Compared to MPC and regular VSM, AMPC provides more consistent and

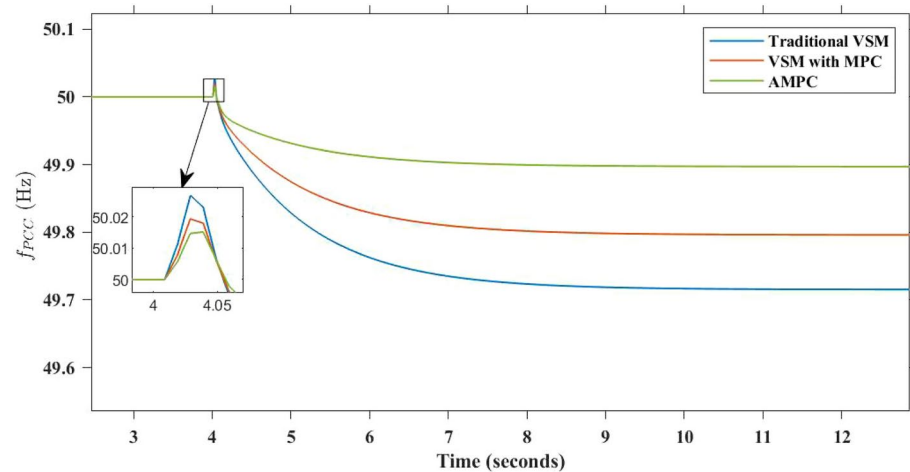


Fig. 15. PCC frequency response during islanding at $t = 4\text{s}$.

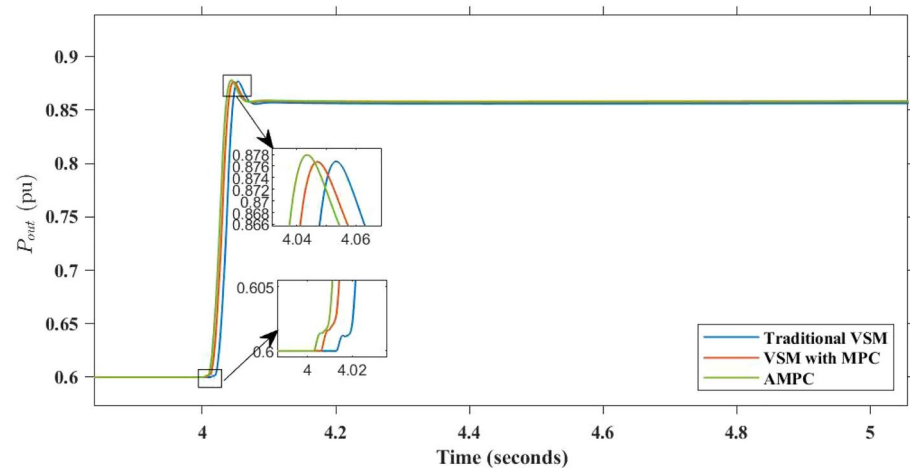


Fig. 16. GFM active power response to frequency variation following island at $t = 4\text{s}$.

Case	Time	Traditional VSM	MPC	AMPC
Frequency OS_{max}	4s	50.027 Hz	50.019 Hz	50.014 Hz
Steady State Frequency after island		49.71 Hz	49.8 Hz	49.89 Hz

Table 12. Summary of frequency behavior at the PCC during Islanding conditions.

stable support for the microgrid during islanding circumstances. Table 12 summarizes the frequency behavior at the PCC during and after island.

Assessment of GFM’s LVRT performance under symmetrical grid faults

In this section, the performance of the GFM is evaluated in terms of its LVRT capability during a severe symmetrical fault, specifically a three-phase line-to-ground (LLLG) fault. The analysis is conducted by simulating the GFM under a grid-connected mode, and graphical results are presented to compare the behavior of the proposed AMPC with traditional MPC and VSM control strategies. This evaluation is essential to determine how well the GFM can stabilize voltage, and frequency, during a grid fault, ensuring its effectiveness in low-inertia systems. The importance of symmetrical fault analysis lies in its direct relevance to the LVRT capability defined by the GC0137 standard, which mandates stringent performance requirements for GFM’s during fault events. Evaluating the GFM’s performance under such conditions is crucial for highlighting the superiority of the developed AMPC, particularly in improving fault ride-through performance and ensuring grid stability.

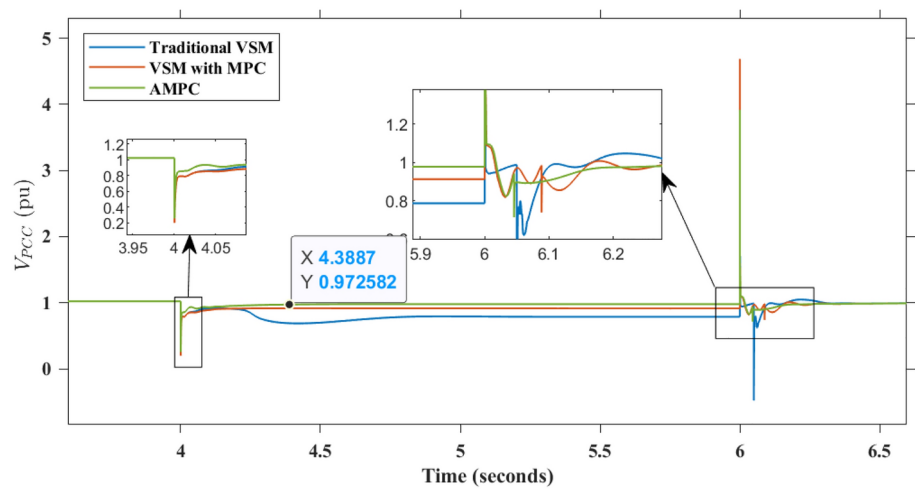


Fig. 17. Voltage sag at the PCC Caused by LLLG Fault from t=4s to t=6s.

Case	Time	Traditional VSM	MPC	AMPC
Voltage dip at the start of fault	4s	0.19 pu	0.19 pu	0.25 pu
Voltage during fault	4s to 6s	0.78 pu	0.91 pu	0.97 pu

Table 13. Summary of voltage fluctuations during LLLG fault across different control strategies.

In this scenario, a three-phase LLLG fault is applied at the local grid side at $t = 4\text{ s}$ and sustained until $t = 6\text{ s}$. The system’s response, particularly the GFM’s ability to stabilize voltage, frequency, and power during this fault, is critical for maintaining grid integrity.

The voltage response at the PCC during the LLLG fault from $t = 4\text{ s}$ to $t = 6\text{ s}$ is shown in Fig. 17. When the fault occurs, the voltage at the PCC drops drastically, reaching as low as 0.1 pu . However, due to the fast-reacting nature of the GFM, which independently manages active and reactive power, the system immediately responds by injecting reactive power to counteract the voltage dip. Notably, the AMPC demonstrates superior voltage stabilization with a minimum voltage dip of 0.25 pu , compared to 0.19 pu in both MPC and VSM control. The voltage rapidly stabilizes at 0.97 pu in the case of AMPC, while MPC and VSM recover to 0.91 pu and 0.78 pu , respectively, throughout the fault duration. After the fault is cleared at $t = 6\text{ s}$, a transient spike occurs due to the combined effect of local grid voltage restoration and the GFM’s reactive power compensation. Although all control schemes show some post-fault oscillations, AMPC achieves the smoothest recovery, settling quickly to the nominal 1 pu voltage, outperforming MPC and VSM, which exhibit more prominent oscillations. This demonstrates the superior efficiency and control precision of the AMPC scheme in managing voltage disturbances. Table 13 summarizes the voltage fluctuations observed during LLLG faults for all control strategies.

Fig. 18 illustrates the reactive power output from the GFM during the fault period. The GFM compensates for the voltage sag by injecting reactive power, with AMPC showing a quicker and higher initial injection. At the onset of the fault, AMPC’s reactive power (Q) peaks at 0.99 pu , outperforming MPC (0.95 pu) and VSM (0.94 pu). During the fault duration, AMPC sustains a reactive power output of 0.96 pu , whereas MPC and VSM settle at 0.82 pu and 0.80 pu , respectively. This enhanced reactive power response allows AMPC to provide more effective voltage support at the PCC, minimizing voltage drops and stabilizing the system. Upon fault clearance at $t = 6\text{ s}$, a dip in reactive power is observed across all control methods, as the system adjusts to the restored grid voltage. However, AMPC once again shows its advanced predictive capabilities by minimizing post-fault oscillations, resulting in a smoother voltage recovery compared to MPC and VSM, as seen in the previous graph.

Since the GFM operation is based on virtual synchronous machine (VSM) control principles, its ability to stabilize frequency is a key measure of performance. Fig. 19 presents the frequency behavior at the PCC during the LLLG fault. As the fault occurs, the frequency deviates from the nominal 50 Hz due to the voltage drop. The AMPC control limits the frequency undershoot to 49.77 Hz , while MPC and VSM experience deeper frequency dips of 49.67 Hz and 49.59 Hz , respectively. Following the undershoot, AMPC shows a fast recovery to the nominal value, whereas MPC experiences some delay, and VSM shows additional dips and slight overshoot before finally stabilizing at 50 Hz . After the fault is cleared at $t=6\text{ s}$, overshoots are observed in all control methods. The VSM overshoots to 50.59 Hz , MPC to 50.33 Hz , and AMPC limits the overshoot to just 50.18 Hz , again showcasing its superior frequency regulation capabilities. The AMPC green line in the graph clearly indicates smoother and quicker stabilization back to 50 Hz , emphasizing the enhanced control offered by the proposed method. Table 14 outlines the frequency behavior observed during LLLG fault.

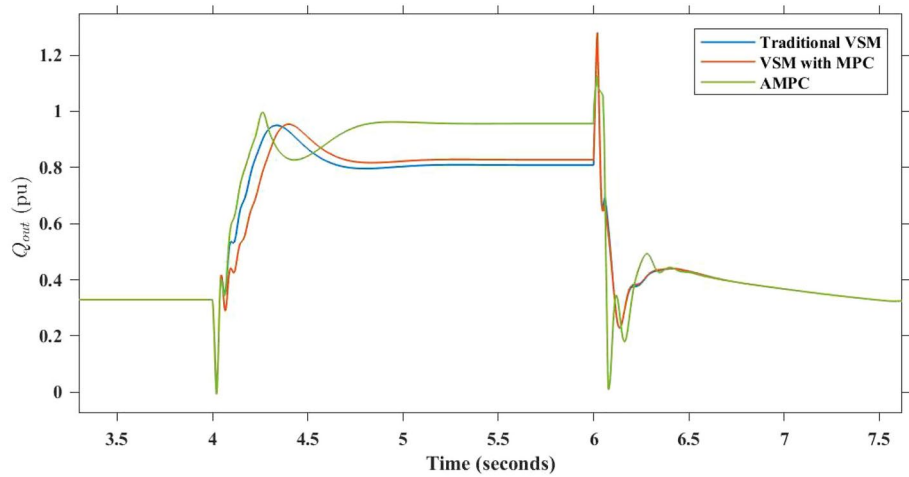


Fig. 18. Reactive power response of the GFM for voltage recovery during LLLG fault event.

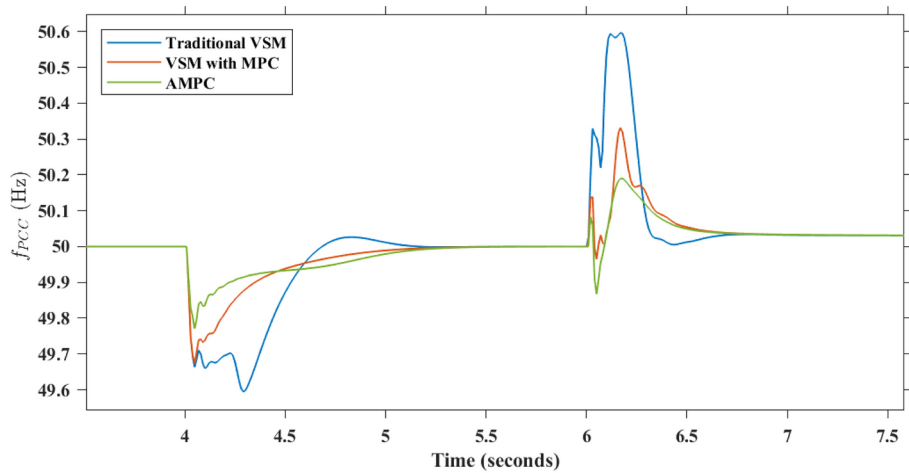


Fig. 19. Frequency stability at the PCC during a severe LLLG fault event (t=4s to t=6s).

Case	Time	Traditional VSM	MPC	AMPC
Frequency US_{max}	4s	49.59 Hz	49.67 Hz	49.77 Hz
Frequency OS_{max}	6s	50.59 Hz	50.33 Hz	50.18 Hz

Table 14. Summary of frequency behavior during LLLG fault across all the Control Strategies.

The GFM compensates for frequency deviations by adjusting its active power output, as shown in Fig. 20. At the beginning of the fault at $t = 4s$, the GFM injects active power to stabilize the frequency at the PCC. The AMPC delivers the highest active power injection, reaching $1.19pu$, compared to $1.15pu$ for MPC and $1.14pu$ for VSM. This higher active power output corresponds to the better frequency stabilization seen in Fig. 19. During the fault, AMPC sustains its active power injection longer, as indicated by the delay in reaching the $0.75pu$ reference value. This prolonged active power support helps regulate frequency more effectively. After the fault is cleared, a dip in active power occurs due to the system balancing the frequency overshoot, with AMPC and VSM reaching a minimum of $-0.74pu$, while MPC dips further to $-0.81pu$. The active power response reflects the GFM’s ability to efficiently regulate frequency during and after fault conditions, with AMPC clearly demonstrating superior handling of frequency deviations.

Throughout this analysis, the GFM’s performance during an LLLG fault demonstrates its capability to meet and exceed the requirements outlined in the GC0137 standard. While the standard specifies a minimum voltage ride-through (VRT) duration of 1 second for voltage drops down to 0% of nominal voltage, the GFM, particularly under AMPC control, maintains operational capability for 2 seconds during the fault. Additionally, the inverter

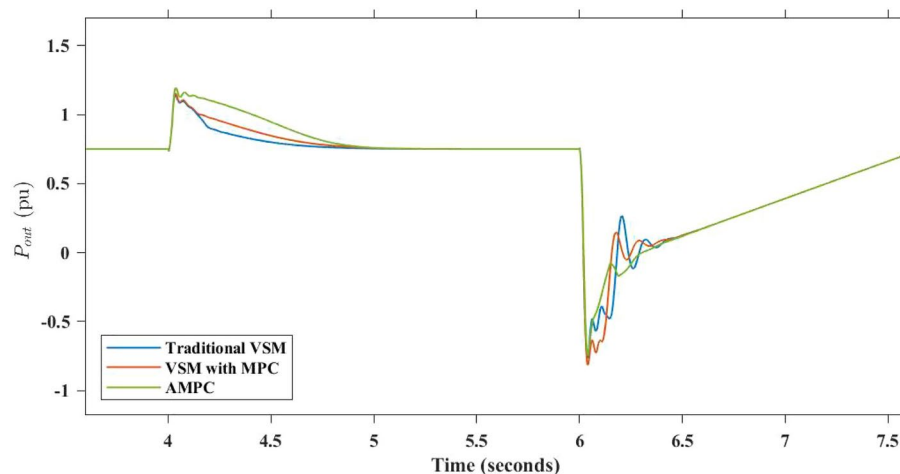


Fig. 20. Active power response of the GFM to mitigate frequency disturbances during LLLG fault.

remains connected even during voltage drops between $0.8pu$ and $0.5pu$, further proving its fault tolerance. These results showcase the enhanced LVRT capability of the proposed AMPC, ensuring greater system reliability and stability under severe fault conditions.

Conclusions and future research

The increased penetration of inverter-interfaced renewable energy resources in modern power grids has significantly reduced system inertia, which is critical for maintaining frequency stability. Among emerging solutions, Grid-Forming Inverters (GFM) have proven pivotal in simulating inertia and enhancing grid stability. This study proposes a novel Adaptive Model Predictive Control (AMPC) framework for GFMs operating in Virtual Synchronous Machine (VSM) mode, designed to optimize performance under dynamic conditions. The AMPC integrates offline reinforcement learning, utilizing the hybrid Crayfish and Self-Adaptive Differential Evolution algorithm (COA-jDE) to minimize the cost function $U_{offline}$, with online Model Predictive Control featuring soft constraints. This approach reduces GFM response time, enhances resilience to grid disturbances, and ensures robust frequency stability while complying with standards such as GC0137 and IEEE 1547.

The efficacy of AMPC was validated against traditional MPC and VSM in a 16 MW DFIG-based wind microgrid under critical scenarios, including grid synchronization and load variations. During grid synchronization with varying reference frequencies, AMPC initiated a response at 4.01s, outperforming MPC (4.06s) and demonstrating superior frequency control efficiency. In cases of load variations and grid voltage phase shifts, AMPC consistently stabilized the PCC more effectively than MPC or VSM. During islanding events, AMPC maintained a steady-state frequency of 49.89 Hz, compared to 49.8 Hz (MPC) and 49.71 Hz (VSM). Similarly, in a low-voltage ride-through (LVRT) scenario with a 2-second LLLG fault, AMPC sustained voltage at 0.97 pu, surpassing MPC (0.91 pu) and VSM (0.78 pu), with minimal post-fault oscillations. These results underscore AMPC's robustness in dynamic environments.

The COA-jDE algorithm's role in minimizing $U_{offline}$ during the offline phase enhances AMPC's computational efficiency, making it suitable for real-time applications. By confining reinforcement learning to offline tuning, AMPC reduces real-time computational burdens, enabling rapid adaptation to frequency deviations without resource overload. The inclusion of soft constraints in the AMPC cost function U_{online} ensures adaptive stability during unexpected deviations, while virtual impedance and current-limiting techniques enhance GFM resilience during faults. The framework's scalability allows deployment across diverse microgrid configurations, ensuring reliable frequency management irrespective of renewable source types or quantities.

While AMPC demonstrates significant advancements, limitations persist, particularly in the offline phase's reliance on precise grid modeling and simulation data. Additionally, scalability in larger networks with potential communication delays requires further validation. Future work will refine these aspects for broader applicability.

Future research will explore distributed and hybrid control strategies for multi-microgrid networks to enhance resilience further. Fine-tuning COA-jDE parameters for complex grid scenarios and extending AMPC to solar PV and PMSG-based systems will test its versatility across renewable energy technologies.

Data availability

All data generated or analyzed during this study are included in this published article.

Received: 31 October 2024; Accepted: 2 May 2025

Published online: 13 May 2025

References

1. Husin, H. et al. A critical review of the integration of renewable energy sources with various technologies. *Protection and control of modern power systems* 6(1), 1–18 (2021).

2. Fernández-Guillamón, A., Gómez-Lázaro, E. & Molina-García, Á. Extensive frequency response and inertia analysis under high renewable energy source integration scenarios: application to the european interconnected power system. *IET Renewable Power Generation* **14**(15), 2885–2896 (2020).
3. Ul-Haq, A., Awan, Yamen S. & Ajum, A. “Frequency control of seig based microgrid during transition from grid connected to island mode,” In *2020 9th International Conference on Industrial Technology and Management (ICITM)*. IEEE, pp. 234–238 (2020).
4. Liu, W. H. et al. Optimal design and sizing of integrated centralized and decentralized energy systems. *Energy Procedia* **105**, 3733–3740 (2017).
5. Qin, B., Wang, M., Zhang, G. & Zhang, Z. Impact of renewable energy penetration rate on power system frequency stability. *Energy Reports* **8**, 997–1003 (2022).
6. Pattabiraman, D., Lasseter, R. & Jahns, T. “Comparison of grid following and grid forming control for a high inverter penetration power system,” In *2018 IEEE Power & Energy Society General Meeting (PESGM)*. IEEE, pp. 1–5 (2018).
7. Li, Y., Gu, Y. & Green, T. C. Revisiting grid-forming and grid-following inverters: A duality theory. *IEEE. Trans. Power. Syst.* **37**(6), 4541–4554 (2022).
8. Ducoin, E. A., Gu, Y., Chaudhuri, B. & Green, T. C. “Analytical design of contributions of grid-forming & grid-following inverters to frequency stability,” *IEEE. Trans. Power. Syst.*, (2024).
9. Naveed, I., Zhao, G., Yamin, Z. & Gul, W. “Steady state performance analysis of dfbg with different magnetizing strategies in a pitch-regulated variable speed wind turbine,” In *2020 4th International Conference on Power and Energy Engineering (ICPEE)*. IEEE, pp. 174–179 (2020).
10. Bikkeli, E., Islam, M. R., Rahman, M. M. & Muttaqi, K. M. State of the art of the techniques for grid forming inverters to solve the challenges of renewable rich power grids. *Energies* **15**(5), 1879 (2022).
11. Rathnayake, D. B. et al. Grid forming inverter modeling, control, and applications. *IEEE Access* **9**, 114 781–114 807 (2021).
12. Mirmohammad, M. & Azad, S. P. Control and stability of grid-forming inverters: A comprehensive review. *Energies* **17**(13), 3186 (2024).
13. Kerdphol, T., Rahman, F. S., Watanabe, M. & Mitani, Y. Robust virtual inertia control of a low inertia microgrid considering frequency measurement effects. *IEEE access* **7**, 57 550–57 560 (2019).
14. Das, T. S., Annakkage, U. D., Muthumuni, D. & Park, I. K. Robust higher order sliding mode control of grid-forming converters with lcl filter in weak grid scenarios for fast frequency support. *IET Generation, Transmission & Distribution*, (2024).
15. Hossain, R. R. et al. Coordinated frequency regulation in grid-forming storage network via safety-consensus. *arXiv preprint arXiv:2410.14877*, (2024).
16. Benhmoudou, Z., Moufid, S., Abbou, A., Ait-Omar, A., Laabassi, H., Chatri, C., Ali, I. H. O. & Bouzekri, H. “A fuzzy logic based controller for synchronization of vsg controlled grid-forming inverters,” in *2023 IEEE PES/IAS PowerAfrica*. IEEE, pp. 1–5 (2023).
17. Palizban, O. & Kauhaniemi, K. Hierarchical control structure in microgrids with distributed generation: Island and grid-connected mode. *Renew. Sustain. Energy. Rev.* **44**, 797–813 (2015).
18. Deng, H. & Fang, J. State-space modeling, stability analysis, and controller design of grid-forming converters with distributed virtual inertia. *Frontiers in Energy Research* **10**, 833387 (2022).
19. Huang, L., Xin, H. & Dorfler, F. H_{∞} — control of grid-connected converters: Design, objectives and decentralized stability certificates. *IEEE. Trans. Smart. Grid.* **11**(5), 3805–3816 (2020).
20. Heydari, R. et al. Model-free predictive control of grid-forming inverters with lcl filters. *IEEE Trans. Power. Electron.* **37**(8), 9200–9211 (2022).
21. Li, Y., Xia, Y., Ni, Y., Peng, Y. & Feng, Q. Transient stability analysis for grid-forming vscs based on nonlinear decoupling method. *Sustainability* **15**(15), 11981 (2023).
22. Qi, X., Zheng, J. & Mei, F. Model predictive control-based load-frequency regulation of grid-forming inverter-based power systems. *Frontiers in Energy Research* **10**, 932788 (2022).
23. Pérez-Ibacache, R. et al. Decentralized model-based predictive control for der units integration in ac microgrids subject to operational and safety constraints. *IEEE Transactions on Power Delivery* **36**(4), 2479–2489 (2020).
24. Gbadega, P. A. & Saha, A. K. Load frequency control of a two-area power system with a stand-alone microgrid based on adaptive model predictive control. *IEEE J. Emerg. Sel. Top. Power. Electron.* **9**(6), 7253–7263 (2020).
25. Wang, L. et al. Adaptive inertia and damping coordination (aidc) control for grid-forming vsg to improve transient stability. *Electronics* **12**(9), 2060 (2023).
26. Liu, Z., Xi, L., Quan, Y., Cheng, C. & Zhu, J. Brain-inspired collaborative automatic generation control with large-scale electric vehicles integration. *IEEE. Trans. Sustain. Energy.* (2024).
27. Liu, Z., Xi, L., Quan, Y. & Cheng, C. Brain-inspired collaborative power dynamic dispatch for the digital twin smart generation system. *IEEE Transactions on Industry Applications* (2024).
28. Kerdphol, T., Watanabe, M., Hongesombut, K. & Mitani, Y. Self-adaptive virtual inertia control-based fuzzy logic to improve frequency stability of microgrid with high renewable penetration. *IEEE Access* **7**, 76 071–76 083 (2019).
29. Gouveia, J., Moreira, C. & Lopes, J. P. Rule-based adaptive control strategy for grid-forming inverters in islanded power systems for improving frequency stability. *Electric Power Systems Research* **197**, 107339 (2021).
30. *GC0137: Grid Code Requirements for Generating Units*, Std., (2022).
31. IEEE, *IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces*, Std., (2018).
32. Babu, V. V., Roselyn, J. P., Nithya, C. & Sundaravadeivel, P. Development of grid-forming and grid-following inverter control in microgrid network ensuring grid stability and frequency response. *Electronics* **13**(10), 1958 (2024).
33. Jia, H., Rao, H., Wen, C. & Mirjalili, S. Crayfish optimization algorithm. *Artificial Intelligence Review* **56**(Suppl 2), 1919–1979 (2023).
34. Deng, W., Xu, J., Song, Y. & Zhao, H. Differential evolution algorithm with wavelet basis function and optimal mutation strategy for complex optimization problem. *Applied Soft Computing* **100**, 106724 (2021).
35. Brest, J., Greiner, S., Boskovic, B., Mernik, M. & Zumer, V. Self-adapting control parameters in differential evolution: A comparative study on numerical benchmark problems. *IEEE. Trans. Evol. Comput.* **10**(6), 646–657 (2006).
36. Yameen, M. Z., Lu, Z., Rao, M. A. A., Mohammad Nasimullah, A. & Younis, W. Improvement of lvrt capability of grid-connected wind-based microgrid using a hybrid goa-pso-tuned statcom for adherence to grid standards. *IET Renewable Power Generation* **18**(15), 3218–3238 (2024).
37. Younis, W. et al. Enhancing load frequency control of interconnected power system using hybrid pso-aha optimizer. *Energies* **17**(16), 3962 (2024).
38. Muisyo, I. N., Muriithi, C. M. & Kamau, S. I. Enhancing low voltage ride through capability of grid connected dfbg based wecs using wca-pso tuned statcom controller. *Heliyon* **8**(8), (2022).
39. Yameen, M. Z. & Lu, Z. Enhancing microgrid stability with a pd-pida controlled statcom using a mgoa for fast-frequency response. *J. Renew. Sustain. Energy.* **17**(1), (2025).
40. Duan, M., Yang, H., Wang, S. & Liu, Y. Self-adaptive dual-strategy differential evolution algorithm. *Plos one* **14**(10), e0222706 (2019).
41. Deng, W. et al. An improved self-adaptive differential evolution algorithm and its application. *Chemometr. Intell. Lab. Syst.* **128**, 66–76 (2013).

42. Wang, H.-B., Ren, X.-N., Li, G.-Q. & Tu, X.-Y. Apdde: self-adaptive parameter dynamics differential evolution algorithm. *Soft computing* **22**, 1313–1333 (2018).
43. Jia, H. et al. Modified crayfish optimization algorithm for solving multiple engineering application problems. *Artificial Intelligence Review* **57**(5), 127 (2024).
44. Fakhouri, H. N., Ishtaiwi, A., Makhadmeh, S. N., Al-Betar, M. A. & Alkhalaileh, M. Novel hybrid crayfish optimization algorithm and self-adaptive differential evolution for solving complex optimization problems. *Symmetry* **16**(7), 927 (2024).
45. Chaib, L., Tadj, M., Choucha, A., Khemili, F. Z. & Attia, E.-F. Improved crayfish optimization algorithm for parameters estimation of photovoltaic models. *Energy. Convers. Manag.* **313**, 118627 (2024).
46. Fan, Q. & Yan, X. Self-adaptive differential evolution algorithm with zoning evolution of control parameters and adaptive mutation strategies. *IEEE Trans. Cybern.* **46**(1), 219–232 (2015).
47. AlKhulaifi, D., AlQahtani, M., AlSadeq, Z., Rahman, A.-U. & Musleh, D. An overview of self-adaptive differential evolution algorithms with mutation strategy. *Mathematical Modelling of Engineering Problems* **9**(4), (2022).
48. Tanabe, R. & Fukunaga, A. Success-history based parameter adaptation for differential evolution. In *2013 IEEE congress on evolutionary computation*. IEEE, pp. 71–78 (2013).

Acknowledgements

We gratefully acknowledge the financial support of Prince Sattam bin Abdulaziz University for our research project, funded under project number PSAU/2025/R/1446.

Author contributions

Conceptualization, M.Z.Y.; methodology, M.Z.Y.; software, M.Z.Y.; validation, M.Z.Y, Z.L and F.F.M; formal analysis, W.Y; investigation, B.A.Z; resources, B.A.Z; data curation, A.K.J; writing original draft preparation, M.Z.Y; supervision, Z.L; project administration, F.F.M; funding acquisition, A.K.J.

Funding

This study is supported via funding from Prince Sattam bin Abdulaziz University project number (PSAU/2025/R/1446).

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to M.Z.Y.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025